

## Review

Synthetic cannabinoids: Epidemiology, pharmacodynamics, and clinical implications<sup>☆</sup>

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## ABSTRACT

**Background:** Synthetic cannabinoids (SC) are a heterogeneous group of compounds developed to probe the endogenous cannabinoid system or as potential therapeutics. Clandestine laboratories subsequently utilized published data to develop SC variations marketed as abusable designer drugs. In the early 2000s, SC became popular as “legal highs” under brand names such as Spice and K2, in part due to their ability to escape detection by standard cannabinoid screening tests. The majority of SC detected in herbal products have greater binding affinity to the cannabinoid CB<sub>1</sub> receptor than does  $\Delta^9$ -tetrahydrocannabinol (THC), the primary psychoactive compound in the cannabis plant, and greater affinity at the CB<sub>1</sub> than the CB<sub>2</sub> receptor. *In vitro* and animal *in vivo* studies show SC pharmacological effects 2–100 times more potent than THC, including analgesic, anti-seizure, weight-loss, anti-inflammatory, and anti-cancer growth effects. SC produce physiological and psychoactive effects similar to THC, but with greater intensity, resulting in medical and psychiatric emergencies. Human adverse effects include nausea and vomiting, shortness of breath or depressed breathing, hypertension, tachycardia, chest pain, muscle twitches, acute renal failure, anxiety, agitation, psychosis, suicidal ideation, and cognitive impairment. Long-term or residual effects are unknown. Due to these public health consequences, many SC are classified as controlled substances. However, frequent structural modification by clandestine laboratories results in a stream of novel SC that may not be legally controlled or detectable by routine laboratory tests.

**Methods:** We present here a comprehensive review, based on a systematic electronic literature search, of SC epidemiology and pharmacology and their clinical implications.

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## 1. Introduction

Synthetic cannabinoids (SC) interact with CB<sub>1</sub> and CB<sub>2</sub> cannabinoid receptors and elicit cannabimimetic effects similar to  $\Delta^9$ -tetrahydrocannabinol (THC), the primary psychoactive constituent in cannabis (Wiley et al., 2013b). SC were developed as research tools to explore the endocannabinoid system and as potential therapeutics (Pertwee, 2006). CB<sub>1</sub> receptors are expressed in the central and peripheral nervous systems, bone, heart, liver, lung, vascular endothelium, and reproductive system (Howlett et al., 2002). CB<sub>2</sub> receptors are primarily in the immune system, but also in the central nervous system at lower levels than CB<sub>1</sub> (Ashton et al., 2006; Onaivi et al., 2008; Van Sickle et al., 2005). SC activate CB<sub>1</sub> receptors, G-protein coupled receptors predominantly located at pre-synaptic terminals. CB<sub>1</sub> receptor activation decreases cellular cyclic adenosine monophosphate (cAMP) levels and elicits cannabimimetic responses (Pertwee, 2010). SC agonists interact with voltage-gated ion channels and inhibit potassium, sodium, and N- and P/Q-type-calcium channels by reducing membrane potentials.

Cyclohexylphenols (CP) were synthesized between 1970 and 1980 with CP55,940 (2-[(1R,2R,5R)-5-Hydroxy-2-(3-hydroxy-propyl)-cyclohexyl]-5-(2-methyl-octan-2-yl)-phenol), commonly utilized to localize cannabinoid receptors (Johnson and Melvin, 1986). Created in Dr. Raphael Mechoulam's laboratory at Hebrew University (HU), Jerusalem, HU-210 [(6aR)-trans-3-(1,1-Dimethylheptyl)-6a,7,10,10a-tetrahydro-1-hydroxy-6,6-dimethyl-6Hdibenzo[b,d]pyran-9-methanol]] is a dibenzopyran, structurally similar to THC, and a highly potent CB<sub>1</sub> and CB<sub>2</sub> agonist (Mechoulam et al., 1990; Howlett et al., 1995; Ovadia et al., 1995; Rodriguez de Fonseca et al., 1995). In the 1990s, aminoalkylindoles such as WIN55,212 [(R)-(+)-[2,3-Dihydro-5-methyl-3-(4-morpholinylmethyl)pyrrolo[1,2,3-de]-1,4-benzoxazin-6-yl]-1-naphthalen-ylmethanone]] were investigated as potentially safer (non-psychotropic) pharmacotherapies (Bell et al., 1991). John W. Huffman (JWH) created the most extensive SC series with chemical structures different from the classical dibenzopyran, but eliciting cannabimimetic effects in animals (Huffman and Dai, 1994). Other SC developed in the last two decades were the AM-series (Alexandros Makriyannis) (Makriyannis and Deng, 2000) and indazole-carboxamide derivatives, e.g., AB-FUBINACA [N-[(2S)-1-Amino-3-methyl-1-oxo-2-butanyl]-1-(4-fluorobenzyl)-1H-indazole-3-carboxamide] (Buchler et al., 2009). To date hundreds of SC were categorized into the following

structural groups (Fig. 1): adamantylindoles, aminoalkylindoles, benzoylindoles, cyclohexylphenols, dibenzopyrans, naphthylindoles, naphthylmethylindoles, naphthylmethylindenes, naphthylpyrroles, phenylacetylindoles, tetramethylcyclopropyl ketone indoles, quinolinyl ester indoles, and indazole carboxamide compounds.

SC, synthesized in clandestine laboratories and sprayed on dried plant materials, were first marketed as legal cannabis alternatives in Europe in the early 2000s (United Nations Office on Drugs and Crime, 2011). SC sold on the Internet, in head shops and convenience stores as Spice and K2 are labeled "not for human consumption." Many SC are Schedule I drugs under the US Controlled Substance Act (US Drug Enforcement Administration, 2014, 2013a,b). As new SC groups are scheduled, more structurally-diverse cannabimimetic compounds emerge, which may not be covered under current regulations.

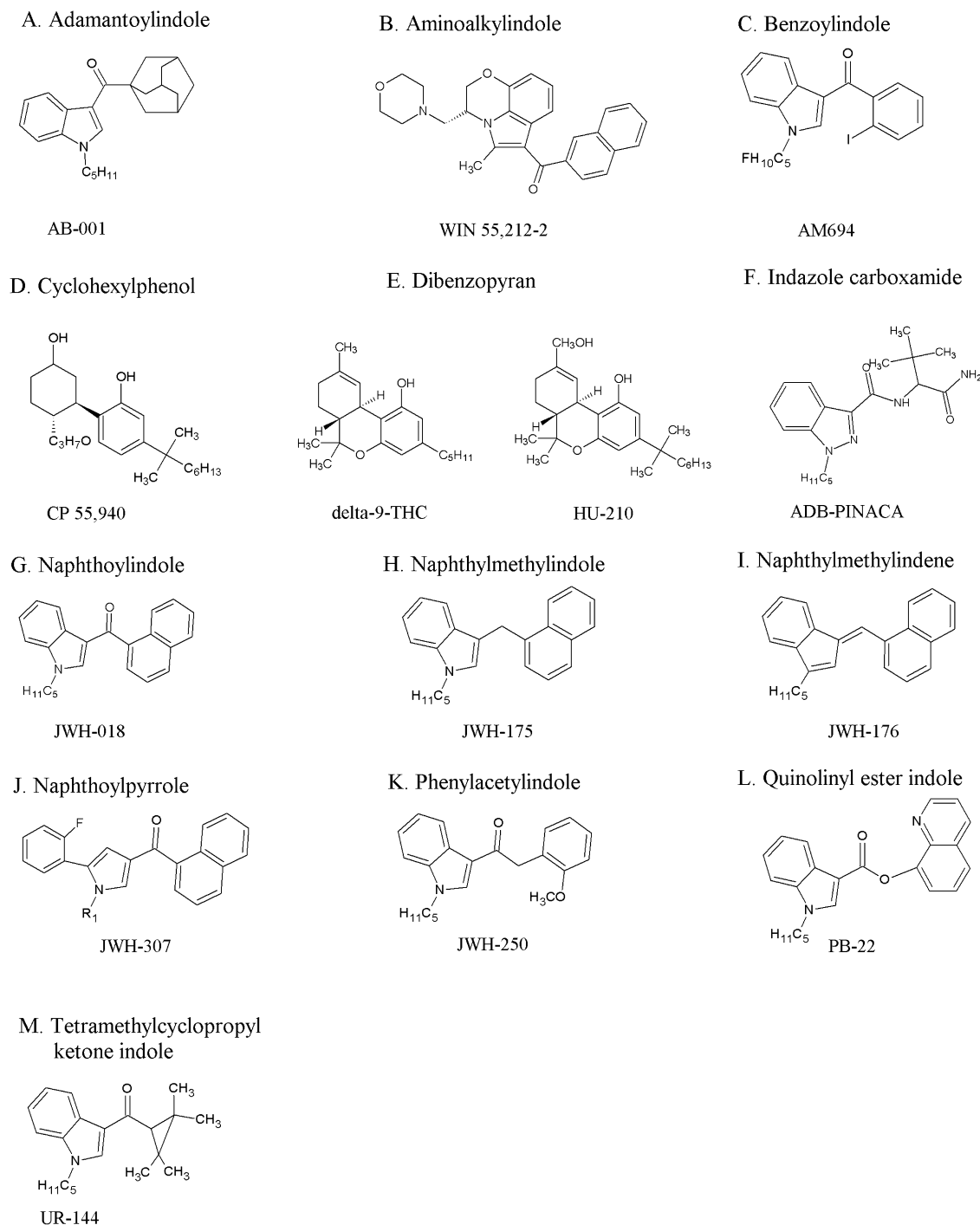
SC popularity are attributed to intense psychoactive effects, lack of detectability in routine urine drug tests, and, until recently, legal status in most jurisdictions (Gunderson et al., 2012; Vandrey et al., 2012; Winstock and Barratt, 2013). Documented serious adverse effects and limited human pharmacology data make SC intake an important public health and safety concern. We present SC epidemiology, pharmacodynamic profiles, and clinical implications, based on a systematic and comprehensive electronic literature review.

## 2. Methods

We conducted a comprehensive literature search of 7 electronic bibliographic databases (PubMed®, Embase™, Web of Science™, Scopus™, Cochrane, Biological Abstracts, and Chemical Abstracts via STN® and SciFinder® platforms) up to December 31, 2013, except that Biological Abstracts (Biosis) and Chemical Abstracts searches ended November 30, 2011. In addition, we expanded our query employing Google and Google Scholar, and hand-searched reference lists of identified articles. We employed database-specific search strategies with multiple keywords (e.g., "synthetic cannabinoids", "JWH-018," "JWH-073," "Spice," and "K2"), utilizing word truncation/wild card symbols and index terms as appropriate for each database (see Supplementary Material Table 1).<sup>1</sup> Each identified article was categorized into the following topics (based on title and abstract review): animal and human pharmacology (pharmacodynamics and pharmacokinetics), chemistry, commercial sources, detection and identification, epidemiology, legal status, receptor interactions, and street use and marketing.

Of 3161 potentially relevant records, 2343 were excluded because they pertained to plant-derived cannabinoids (e.g., cannabinol, cannabidiol, nabiximols

<sup>1</sup> Supplementary material can be found by accessing the online version of this paper. Please see Appendix A.



**Fig. 1.** Structural classes of synthetic cannabinoids. AB-001 [1-Pentyl-3-(1-adamantoyl)indole], ADB-PINACA [N-(1-Amino-3,3-dimethyl-1-oxobutan-2-yl)-1-(4-fluorobenzyl)-1H-indazole-3-carboxamide], AM694 [1-(5-Fluoropentyl)-1H-indol-3-yl](2-iodophenyl)-methanone, delta-9-THC (delta-9-tetrahydrocannabinol), HU-210 [3-(1,1'-Dimethylheptyl)-6aR,7,10,10aR-tetrahydro-1-hydroxy-6,6-dimethyl-6H-dibenzo[b,d]pyran-9-methanol], JWH-018 [(1-Pentyl-1H-indol-3-yl)-1-naphthalenylmethanone], JWH-175 [3-(1-Naphthalenylmethyl)-1-pentyl-1H-indole], JWH-176 [1-([(1E)-3-Pentylinden-1-ylidene]methyl)naphthalene], JWH-250 [(2-Methoxyphenyl)-1-(1-pentylindol-3-yl)ethanone], JWH-307 [5-(2-Fluorophenyl)-1-pentylpyrrol-3-yl-naphthalen-1-ylmethanone], PB-22 (1-Pentyl-1H-indole-3-carboxylic acid 8-quinolinyl ester), UR-144 [(1-Pentylindol-3-yl)-(2,2,3,3-tetramethylcyclopropyl)methanone].

[Sativex®]), synthetic THC (dronabinol [Marinol®], nabilone [Cesamet®]), or endogenous cannabinoids (e.g., anandamide, 2-arachidonoylglycerol), or were not English language. Articles ( $n=818$ ) related to SC analytical methods, chemical synthesis, legal status, pharmacokinetics, and street use and marketing will be reviewed elsewhere; this review covers SC epidemiology (9), animal and human pharmacodynamics (338), and receptor interactions (42). Articles older than 10 years were included only if the study was not replicated more recently, yielding 215 articles included in this review.

### 3. Results

#### 3.1. Epidemiology

We identified 9 SC epidemiological studies, none population- or community-based. Two worldwide surveys of self-selected convenience samples were conducted between 2011 and 2012. The

first, collecting data online between January 4th and February 7th, 2011 recruited participants primarily from Internet drug forums providing SC information (Vandrey et al., 2012). Of 391 participants, 168 met inclusion criteria (self-reported SC intake, English speaking, and  $\geq 18$  years), representing 13 countries and 42 US states. The majority were Caucasian (90%), single (67%), men (83%), with at least high school education (96%), with mean first SC use at 26 years. Almost all (92%) reported alcohol and 84% cannabis intake, both frequently combined with SC. The most common route of administration was smoked (via water pipe/bong, cigarette, blunt or pipe); however, oral, rectal and vaporized administration also were mentioned. Curiosity was the primary reason (78%) for SC intake, while 58% favored the drugs' effects. About one-third sought intoxication while avoiding a positive cannabinoid test. SC abuse was reported by 37%, dependence 15%, and withdrawal 15%. Discontinued SC intake produced similar cannabis withdrawal symptoms including headaches, anxiety/nervousness, depression, irritability.

The second worldwide anonymous online survey collected data between November 23 and December 21, 2011, yielded 14,966 participants, two-thirds men, median age 26 years, of whom 17% ( $n = 2513$ ) reported SC intake (Winstock and Barratt, 2013). Of those using SC within the last year ( $n = 980$ ), 98% also used cannabis, and commonly other drugs. Although SC onset of effects was faster than for cannabis, 92.8% ( $n = 887$ ) preferred cannabis to SC due to the latter's undesirable effects. However, 7.2% reported preference for SC over cannabis for reasons such as accessibility, cost, non-detection, and effects.

Several single-country surveys of self-selected convenience samples were conducted. An anonymous email survey ( $n = 852$ ) at the University of Florida in September, 2010 found that 8.1% reported lifetime SC use and 68% were male and 32% female. Smoking in a cigarette was most frequent (88%), with 25% consuming with a hookah and/or joint (Hu et al., 2011). A majority (91.3%) also smoked cannabis. There was no significant association between SC use and race, marital status, or campus living arrangement.

Between December, 2011 and January, 2012, 316 Australian SC users ( $\geq 18$  years, 77% male, 86% high school graduates, and 30% college graduates) completed an online survey (Barratt et al., 2013). The majority (78%) were employed, 19% were students, and <10% unemployed (or actively looking for work). About 10% sought aid for illicit drug and 3% for alcohol-related problems. Almost all (94%) reported SC intake within the past year, 45% within 30 days. SC daily use was reported by 7% and weekly intake by >33%. 96% reported lifetime cannabis intake, 94% within the last year, and 61% the past month. Polydrug use was reported by 64% and 27 different SC brands were consumed.

A field-based survey interviewed 1740 adults between May and October, 2012 at clubs in New York City (Kelly et al., 2013). Mean age was 26.4 (range 18–40) years, with 55.2% male, 35.5% homosexual, 61% Caucasian, 15% African-American, 15% Latino, 6.0% Asian, and 12% mixed race. Of those reporting past year SC intake (8.2%), 41.2% were heterosexual men and 17.4% lesbian and bisexual women. SC prevalence in this US sample was lower than self-reported prevalence (12.6%) among those participating ( $n = 2700$ ) in an online survey in UK dance clubs between November 17 and 30, 2009 (Winstock et al., 2011).

The annual Monitoring the Future survey of approximately 50,000 US high school students included questions related to lifetime and past year SC intake in 2011–2013 (O'Malley et al., 2014). For 2013, 25.8% reported past-year cannabis intake and 6.4% SC intake. In 2012, students reported 24.7% cannabis and 8.0% SC consumption, with 14.8% and 23.5% of 12th graders perceiving cannabis and SC intake (once or twice), respectively, as less harmful than other drugs. By 2013, SC perceived harmfulness increased to 25.9% (not significant), while perceived harmfulness of most other

illicit drugs, aside from synthetic cathinones (59.5%), did not significantly change.

SC possession and use were prohibited for US military personnel as early as 2010 (Cole, 2011; Gould, 2010, 2013; Live, 2011; Loeffler and Hurst, 2011). Between March, 2011 and March, 2012, the Armed Forces Medical Examiner System received 1635 urine samples from service members suspected of SC intake (Brantley, 2012). Of these, 1148 (70.2%) specimens confirmed positive for SC. An Army Substance Abuse Program 2012 SC prevalence study of 10,000 randomly collected urine specimens, previously screened negative for routinely-tested drugs, reported 2.5% positivity rate (Gould, 2013). The 2011 Health Related Behaviors Among Active Duty Military Personnel web-based survey reported a 1.1% past month SC use among 39,877 non-deployed member respondents, which was higher than cannabis (0.9%) (Barlas et al., 2013). Though some service members claimed that SC sold in convenience stores were perceived as legal (Effron, 2013; Logico, 2012), military officials believed that limited SC testing was a major factor in their illicit use (Cole, 2011; Gould, 2013). SC are now included in routine military drug testing (Army Center for Substance Abuse Program, 2013).

In 2012, the World Anti-Doping Agency (WADA) reported 8 samples positive for JWH-018 [naphthalen-1-yl-(1-pentylindol-3-yl)methanone] ( $n = 6$ ) and JWH-073 [naphthalen-1-yl-(1-butyldiol-3-yl)methanone] ( $n = 2$ ) among 4500 samples positive for prohibited substances under the Anti-Doping Administration and Management System (World Anti-Doping Agency, 2012). HU-210 or THC-like cannabinoids were identified in the 2010 Prohibited List (World Anti-Doping Agency, 2010), which further expanded in 2011 to include "Spice" cannabimimetics (World Anti-Doping Agency, 2011).

In summary, SC appealed to young cannabis and polydrug users. US high school students perceive SC as safer than other drugs of abuse. Military personnel and athletes smoke SC to avoid a positive drug test. However, most SC smokers preferred cannabis due to SC' negative effects.

### 3.2. Animal pharmacodynamics

There were 265 peer-reviewed journal articles published between 1989 and 2013 relating to SC animal pharmacodynamics. The most frequently studied SC were the aminoalkylindole WIN55,212-2 (134 studies), cyclohexylphenol CP55,940 (54), and HU-210 (39). We identified few studies with the naphthoylindoles (JWH-018 and analogs) and the newer SC structural families that now dominate the market.

**3.2.1. Physiological and behavioral effects.** Acute SC intraperitoneal (i.p.), intravenous (i.v.), or oral (p.o.) administration produced dose-dependent physiological effects similar to THC in rodents and non-human primates, such as anti-emesis, cannabinoid tetrad (analgesia, catalepsy, hypomotility, hypothermia), and hypotension (see Table 1). Some SC act as anti-epileptics or enhance anti-convulsant drug actions in rodents. SC activity at the CB<sub>2</sub> receptor mediates anti-cancer, anti-inflammatory, anti-oxidative cardio-protective, and immunosuppressive properties (Table 1).

Few SC studies evaluated effects on more complex animal behavior. Chronic CP55,940 (0.15–0.3 mg/kg, i.p.) exposure in adolescent rats impaired short-term and working memory as adults, while chronic exposure in adulthood had no long-lasting effect (Renard et al., 2013). Acute administration of 1.2 mg/kg i.p. WIN55,212-2 to adolescent and adult rats affected object and social discrimination; rats continuously exposed to WIN55,212-2 since puberty showed persistent discrimination deficits as adults (Schneider et al., 2008). Behavioral deficits (socialization, self-grooming) were more pronounced in WIN55,212-2 (1.2 mg/kg

**Table 1**  
Synthetic cannabinoids (SC) pharmacodynamic evaluation and effects in animals.

| Animal | Drug                                                    | Dose                                                                      | Route | Exposure type    | ED <sub>50</sub> | Measured effects                     | Results                                                                                                                                                                                                                                                                                                                                                                               | Reference                                    |
|--------|---------------------------------------------------------|---------------------------------------------------------------------------|-------|------------------|------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Rat    | HU-210                                                  | 1, 5, 25 µg/kg                                                            | p.o.  | Chronic          | Not evaluated    | Prenatal exposure                    | HU-210 exposure had no effect on mother's weight gain, food, & water intake. Exposed pups size & weight also did not vary from control group; however, pups exposed to 1 µg/kg had higher spleen weight than other HU-210-exposed group & control.                                                                                                                                    | <a href="#">del Arco et al. (2000)</a>       |
| Rat    | HU-210,RIM                                              | 4–100 µg/kg                                                               | s.c.  | Acute            | Not evaluated    | Drug relapse                         | HU-210 dose-dependently re-instated drug-seeking behavior in COC-exposed rats (abstained for 2 wk) which was attenuated with RIM pre-treatment                                                                                                                                                                                                                                        | <a href="#">De Vries et al. (2001)</a>       |
| Mouse  | R(+)- & S(–) WIN55,212-2                                | 5, 20 mg/kg                                                               | i.p.  | Chronic          | Not evaluated    | Disease treatment                    | High dose R(+)-WIN55,212-2 on day of TMEV infection delayed onset of disease & reduced severity compared to control & S(–)-WIN55,212-2.                                                                                                                                                                                                                                               | <a href="#">Croxford and Miller (2003)</a>   |
| Mouse  | WIN55,212-2, ACEA, JWH-015                              | mg/kg: 2.5–5 WIN55,212-2, 1.25–2.5 ACEA, 0.6–1.2 JWH-015                  | i.p.  | Chronic          | Not evaluated    | Disease treatment                    | SC improved motor functions in multiple sclerosis animal model & reduced microglial activation in TMEV-infected animals                                                                                                                                                                                                                                                               | <a href="#">Arevalo-Martin et al. (2003)</a> |
| Rat    | WIN55,212-2                                             | 2.0–8.0 mg/kg                                                             | i.p.  | Subchronic       | Not evaluated    | Drug cross-tolerance                 | Adolescent subchronic WIN55,212-2 exposure altered DA neurons responsivity in animals when challenged with COC, MOR or AMP compared to control or adult animals exposed under the same conditions.                                                                                                                                                                                    | <a href="#">Pistis et al. (2004)</a>         |
| Mouse  | HU-320                                                  | 1, 2 mg/kg                                                                | i.p.  | Acute            | Not evaluated    | Disease treatment                    | HU-320 possessed anti-inflammatory & immunosuppressive properties in a rheumatoid arthritis animal model without its psychoactive effects.                                                                                                                                                                                                                                            | <a href="#">Sumariwalla et al. (2004)</a>    |
| Rat    | HU-210, AM404, AM251, oleamide, desipramine (+ control) | 5, 25 µg/kg HU-210, 1& 5 mg/kg AM404, 5 mg/kg AM251, 10 mg/kg desipramine | i.p.  | Acute            | Not evaluated    | Immobility                           | Desipramine, AM404, HU-210, & oleamide dose-dependently reduced mobility (forced swim test) in animal, which was blocked by AM251 pre-treatment.                                                                                                                                                                                                                                      | <a href="#">Hill and Gorzalka (2005)</a>     |
| Rat    | HU-210                                                  | 0.1 mg/kg                                                                 | i.p.  | Chronic          | Not evaluated    | Immobility, Neurogenesis             | Acute HU-210 decreased mobility in adult rats; chronic HU-210 exposure promoted hippocampal neurogenesis.                                                                                                                                                                                                                                                                             | <a href="#">Jiang et al. (2005)</a>          |
| Rat    | AM411                                                   | 0.3–5.0 mg/kg                                                             | i.p.  | Acute            | Not evaluated    | Cannabinoid tetrad                   | AM411 dose-dependently produced cannabinoid tetrad, impaired animal in operant lever pressing, & reduced animal's activity in open field test. Effects were attenuated by AM251, CB1 antagonist.                                                                                                                                                                                      | <a href="#">McLaughlin et al. (2005)</a>     |
| Rat    | CP55,940, MOR                                           | mg/kg: 0.2 WIN55,212-2, MOR                                               | i.p.  | Acute Subchronic | Not evaluated    | Analgesia, Drug cross-tolerance      | Acute CP55,940 & MOR exposure, when administered alone, did not significantly elicit analgesia. However, administered together significantly reduced pain in animal. Acute CP55,950 (0.2 mg/kg) administration to MOR-tolerant rat produced analgesia, which was not observed in control group. Acute MOR administration to CP55,940-tolerant rat did not significantly reduced pain. | <a href="#">Vigano et al. (2005)</a>         |
| Rat    | HU-210, 8-OH-DPAT, DOI                                  | mg/kg: 0.1 HU-210, 0.3 8-OH-DPAT, 1 DOI                                   | i.p.  | Chronic          | Not evaluated    | Drug cross-tolerance                 | Chronic HU-210 treatment increased frequency of DOI-induced wet-dog shakes compared to control (non-HU-210 exposed); Hypothermia-induced 8-OH-DPAT did not differ between HU-210 pre-treated animals & control                                                                                                                                                                        | <a href="#">Hill et al. (2006)</a>           |
| Rat    | HU-210                                                  | 5, 100 µg/kg                                                              | i.p.  | Chronic          | Not evaluated    | Increased stress hormones & response | Animals exposed to 100 µg/kg HU-210 had significantly higher serum corticosterone levels, which correlated to increased stress responsivity in adult animals compared to low dose or vehicle exposed group.                                                                                                                                                                           | <a href="#">Hill and Gorzalka (2006)</a>     |
| Rat    | WIN55,212-2, RIM, AM1387                                | mg/kg: 1–5.6 WIN55,212-2 & RIM, 3 & 10 AM1387                             | i.p.  | Acute            | Not evaluated    | Immobility                           | WIN55,212-2 dose-dependently reduced locomotor & grooming activities, which were attenuated by RIM & AM1387 pre-treatment                                                                                                                                                                                                                                                             | <a href="#">Järbe et al. (2006)</a>          |



|       |                                          |                                                             |           |         |                                                                      |                                               |                                                                                                                                                                                                                                                                                                                                                                                           |                           |
|-------|------------------------------------------|-------------------------------------------------------------|-----------|---------|----------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Mouse | WIN55,212-2, ETOH, 8-OH-DPAT             | 0–10% ETOH, 0.5 mg/kg WIN55,212-2, 0.25 & 1 mg/kg 8-OH-DPAT | i.p. s.c. | Chronic | Not evaluated                                                        | Increased ETOH intake in binge-drinking model | C57BL/6J mice progressively increased ETOH intake than DBA/2J mice with increase body weight for both strains. Chronic WIN55,212-2 treatment did not significantly change ETOH intake while chronic 8-OH-DPAT (5HT1A receptor agonist) decreased total ETOH intake in C57BL/6J mice. WIN 55,212-2 chronic treatment increased ETOH in DBA/2J mice, which was not observed with 8-OH-DPAT. | Kelai et al. (2006)       |
| Rat   | WIN55,212-2, AM251, SR144528             | mg/kg: 0.06–0.25 WIN55,212-2, 1 AM251 & SR144528            | i.v.      | Acute   | Not evaluated                                                        | Analgesia                                     | WIN55,212-2 dose-dependently attenuated hyperalgesia in a neuropathic pain animal model, which was blocked only by AM251 (CB <sub>1</sub> antagonist).                                                                                                                                                                                                                                    | Liu and Walker (2006)     |
| Rat   | WIN55,212-2                              | 1.5 mg/kg                                                   | i.p. s.c. | Chronic | Not evaluated                                                        | Immobility, Disease treatment                 | WIN55,212-2 dose-dependently reduced locomotor activity & attenuated acute paraoxon-induced cholinergic toxicity; however, increased movement was observed when paraoxon was given after 7d of WIN55,212-2 daily exposure                                                                                                                                                                 | Nallapaneni et al. (2006) |
| Rat   | WIN55,212-2, quinolinic acid (QA), AM251 | 5, 10 µM WIN55,212-2, 5 mM QA. 0.5 mg/kg AM251              | m.d.      | Acute   | Not evaluated                                                        | Disease treatment                             | WIN55,212-2 pre-treatment prevented extracellular level increased in the striatum induced by QA. Effect was blocked by AM251.                                                                                                                                                                                                                                                             | Pintor et al. (2006)      |
| Rat   | WIN55,212-2                              | 0.5 mg/kg                                                   | s.c.      | Chronic | Not evaluated                                                        | Prenatal exposure                             | Exposure to WIN55,212-2 had no effect on pre/postnatal progression & development, however, changes in GAD & GABA immunocross-reactivities in the GABAergic neuronal system were observed.                                                                                                                                                                                                 | Benagiano et al. (2007)   |
| Rat   | WIN55,212-2, THC                         | mg/kg: 0.5 WIN55,212-2, 5 THC                               | s.c.      | Chronic | Not evaluated                                                        | Prenatal exposure                             | Prenatal WIN55,212-2 exposure reduced extracellular glutamate levels in front cerebral cortex of adolescent rats similar to THC & increased glutamate uptake by overexpression of glutamate transporter subtypes (GLT1 & EAAC1)                                                                                                                                                           | Castaldo et al. (2007)    |
| Mouse | HU-210                                   | 0.1 mg/kg                                                   | i.p.      | Chronic | Not evaluated                                                        | Disease treatment                             | HU-210 improved performance of diabetic mice, but had no effect on hyperglycemic index or animal's weight.                                                                                                                                                                                                                                                                                | Dagon et al. (2007)       |
| Rat   | WIN55,212-2, HU-210 (+ control)          | 0.3–5 mg/kg WIN55,212-2, 10 µg/kg HU210                     | i.p.      | Acute   | 3.4–7.6 mg/kg WIN55,212-2 (thermal pain); 0.7–2.0 mg/kg (mechanical) | Analgesia                                     | WIN55,212-2 dose-dependently attenuated thermal & mechanical pain in trigeminal neuropathic animal pain model, which was blocked by CB <sub>1</sub> antagonist only.                                                                                                                                                                                                                      | Liang et al. (2007)       |
| Rat   | WIN55,212-2, bupivacaine                 | 0.3–10 µg WIN 55,212-2; 1–100 µg bupivacaine                | i.t.      | Acute   | 0.12–0.89 µg WIN 55,212-2; 0.02–0.1 µg WIN 55,212-2 + bupivacaine    | Analgesia                                     | WIN55,212-2 possess analgesic properties & effectiveness increased with co-administration of bupivacaine in rat formalin-induced pain model                                                                                                                                                                                                                                               | Kang et al. (2007)        |
| Rat   | WIN55,212-2                              | 3 mg/kg                                                     | i.p.      | Chronic | Not evaluated                                                        | Anxiogenic                                    | Repeated WIN55,212-2 elicited anxiogenic effects in animal that correlated with increased tyrosine hydroxylase expression in the locus coeruleus.                                                                                                                                                                                                                                         | Page et al. (2007)        |
| Rat   | WIN55,212-2                              | 0.25, 1.25 mg/kg                                            | i.p.      | Acute   | Not evaluated                                                        | Hyperactivity                                 | WIN55,212-2 increased locomotor activity in suitable genetic ADHD animal model while it reduced in control group                                                                                                                                                                                                                                                                          | Pandolfo et al. (2007)    |
| Rat   | WIN55,212-2, RIM                         | 60, 90 µg/kg WIN55,212-2, 50 pg/kg RIM                      | i.v.      | Acute   | Not evaluated                                                        | Analgesia                                     | Co-administration of WIN55,212-2 & low dose RIM produced & extended antinociception in animal beyond 7 d. Animal exposed to WIN55,212-2 only developed tolerance by after a week.                                                                                                                                                                                                         | Paquette et al. (2007)    |
| Mouse | WIN55,212-2                              | 2, 4 mg/kg                                                  | i.p.      | Acute   | Not evaluated                                                        | Analgesia, Catalepsy                          | WIN55,212-2 attenuated pain in ovariectomized mice, which was blocked by 10 mg/kg estrogen. Progesterone (25 mg/kg) administration facilitated cataleptic effects from WIN55,212-2 (2 mg/kg) administration, not estradiol.                                                                                                                                                               | Anaraki et al. (2008)     |

**Table 1** (Continued)

| Animal | Drug                                               | Dose                                                                                | Route       | Exposure type | ED <sub>50</sub>                                            | Measured effects                                     | Results                                                                                                                                                                                                                                                                                                                                                                            | Reference                                    |
|--------|----------------------------------------------------|-------------------------------------------------------------------------------------|-------------|---------------|-------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Rat    | CP55,940                                           | 0.4 mg/kg                                                                           | i.p.        | Chronic       | Not evaluated                                               | Drug desensitization                                 | CP55,940 peridolescent exposure increased morphine (1 mg/kg) self-administration & changed $\mu$ -opioid receptor density in adult rats, more in male than female rats.                                                                                                                                                                                                            | <a href="#">Biscaia et al. (2008)</a>        |
| Rat    | WIN55,212-2, APDC & L-AP4                          | 3–30 $\mu$ g per 10 $\mu$ L WIN55,212-2, 100 nM APDC, 30 nM L-AP4                   | i.c.        | Acute         | Not evaluated                                               | Analgesia                                            | Intracisternal WIN55,212-2 injections at 30 $\mu$ g produced analgesia effects in formalin-induce TMJ-pain animal model. Co-administration of 3 $\mu$ g WIN55,212-2 & 100 nmol of metabotropic glutamate receptors agonists APDC & L-AP4 induced analgesic effects.                                                                                                                | <a href="#">Lee et al. (2008)</a>            |
| Mouse  | WIN55,212-2, CP55,940, THC (+ control)             | mg/kg: 0.1, 3.2 SC, 10 THC                                                          | i.p.        | Acute         | mg/kg: 0.00–0.06 CP55,940, 0.3–0.7 WIN55,212-2, 1.8–3.7 THC | Drug discrimination                                  | SC substituted for THC in drug discrimination study with potency order: CP55,940 > WIN55,212-2 > THC.                                                                                                                                                                                                                                                                              | <a href="#">McMahon et al. (2008)</a>        |
| Rat    | WIN55,212-2                                        | 1.2 mg/kg                                                                           | i.p.        | Chronic       | Not evaluated                                               | Behavioral deficits from adolescent exposure         | Chronic WIN55,212-2 exposure induced prominent object/social recognition & social behavioral deficits in adolescent-exposed than adult rats                                                                                                                                                                                                                                        | <a href="#">Schneider et al. (2008)</a>      |
| Rat    | WIN55,212-2                                        | 3 mg/kg                                                                             | i.p. i.t.   | Acute         | Not evaluated                                               | Analgesia                                            | WIN55,212-2 acute administration attenuated induced-pain in mouse                                                                                                                                                                                                                                                                                                                  | <a href="#">Dableh et al. (2009)</a>         |
| Mouse  | WIN55,212-2, ETOH                                  | 0.5–2 mg/kg WIN55,212-2, 20% ETOH                                                   | i.p. p.o.   | Acute         | Not evaluated                                               | Increased ETOH intake in binge-drinking animal model | Low WIN55,212-2 (0.5 mg/kg) administration increased ETOH intake, but decreased with higher dose (1 & 2 mg/kg) in a binge-drinking (C57BL/6J B6) animal model                                                                                                                                                                                                                      | <a href="#">Linsendardt and Boehm (2009)</a> |
| Rat    | WIN55,212-2                                        | 1, 2, 5 mg/kg                                                                       | i.c.v. i.p. | Acute         | Not evaluated                                               | Increased appetite                                   | WIN55,212-2 at 1 & 2 mg/kg increased rat's appetite, but reduced at 5 mg/kg similar to AM251                                                                                                                                                                                                                                                                                       | <a href="#">Merroun et al. (2009)</a>        |
| Mouse  | WIN55,212, JWH-133, AM251, SR144528                | 5 mg/kg                                                                             | peri-tumor  | Chronic       | Not evaluated                                               | Disease treatment                                    | SC reduced tumor volume in animals injected with SC for 4 wk                                                                                                                                                                                                                                                                                                                       | <a href="#">Qamri et al. (2009)</a>          |
| Rat    | WIN55,212-2                                        | 1.2 mg/kg                                                                           | i.p.        | Chronic       | Not evaluated                                               | Behavioral deficits from adolescent exposure         | cFos protein expression in brain of adult rats altered after chronic WIN55-212-2 pubertal exposure; PPI deficits (reduced) also observed compared to control group                                                                                                                                                                                                                 | <a href="#">Wegener and Koch (2009)</a>      |
| Rat    | CP55,940, THC, nicotine, AM251                     | mg/kg: 0.4 nicotine, 3.0 THC & AM251, 0.1–0.4 CP55,049                              | i.p.        | Acute         | Not evaluated                                               | Immobility, Drug cross-tolerance                     | Subchronic nicotine treatment in adolescent rats significantly decreased locomotor activity after THC & CP55,940 administrations compared to control (vehicle) group. This was not observed in adult rats under the same conditions. Moreover, CB1 receptor binding significantly increased in the medial prefrontal cortex of adolescent rats subchronically exposed to nicotine. | <a href="#">Werling et al. (2009)</a>        |
| Rat    | CP55,940                                           | 10–50 $\mu$ g/kg                                                                    | i.p.        | Acute         | Not evaluated                                               | Anxiogenic                                           | CP55,940 dose-dependently induced anxiety in Wistar rats & not Lewis rats                                                                                                                                                                                                                                                                                                          | <a href="#">Arnold et al. (2010)</a>         |
| Rat    | WIN55,212-2, MOR, naloxone, indomethacin, ketamine | mg/kg: 0.5. 1 WIN55,212-2, 2–10 MOR, 1 or 2 naloxone, 2.5 or 5 indomethacin, 25, 50 | i.p.        | Acute         | Not evaluated                                               | Analgesia                                            | WIN55,212-2 attenuated formalin-induced orofacial pain more potent than MOR, indomethacin, & ketamine.                                                                                                                                                                                                                                                                             | <a href="#">Burgos et al. (2010)</a>         |

|            |                                     |                                           |           |                          |               |                                                                |                                                                                                                                                                                                                                                                          |                                                                                        |
|------------|-------------------------------------|-------------------------------------------|-----------|--------------------------|---------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Mouse      | HU-210                              | 10, 50 µg/kg                              | i.p.      | Chronic                  | Not evaluated | Disease treatment                                              | HU-210 did not improve water-maze performance in Alzheimer's disease animal model; moreover, it had no effect on amyloid β protein formation nor it enhances neurogenesis in the brain.                                                                                  | <a href="#">Chen et al. (2010)</a>                                                     |
| Rat        | HU-210                              | 25–100 µg/kg                              | i.p.      | Acute Subchronic-Chronic | Not evaluated | Weight loss/gain                                               | Weight gain was slowest with adult rats exposed 100 µg/kg HU-210, but dose-dependently improved after subchronic (4 d) exposure. Acute, subchronic & chronic HU-210 treatment reduced or downregulated CB1 receptor binding, more prominent in adult than adolescent rat | <a href="#">Dalton et al. (2009)</a> and <a href="#">Dalton and Zavitsanou (2010a)</a> |
| Rat        | HU-210                              | 25–100 µg/kg                              | i.p.      | Acute Sub-chronicChronic | Not evaluated | Increased dopamine receptor density                            | Dopamine (D2) receptor density increased in lateral caudate putamen & olfactory tubercle brain area of adult rats after 14 day HU-210 100 µg/kg exposure.                                                                                                                | <a href="#">Dalton and Zavitsanou (2010b)</a>                                          |
| Rat        | CP55,940                            | 20 & 40 µg/kg                             | i.p.      | Acute                    | Not evaluated | Altered sexual behavior                                        | CP55,940 dose-dependently suppressed sexual motivation in female rats, previously exposed to sexual hormones                                                                                                                                                             | <a href="#">Lopez et al. (2010)</a>                                                    |
| Mouse      | WIN55,212-2, MDMA                   | mg/kg: 1.25–5 MDMA, 0.1 & 0.5 WIN55,212-2 | i.p.      | Acute                    | Not evaluated | Drug relapse                                                   | WIN55,212-2 & MDMA induced conditioned place preference (CPP). WIN55,212-2 0.5 mg/kg reinstated MDMA-induced CPP.                                                                                                                                                        | <a href="#">Manzanedo et al. (2010)</a>                                                |
| Guinea pig | WIN55,212-2, MOR                    | mg/kg: 10–80 MOR, 6 WIN55,212-2           | s.c. i.p. | Chronic                  | Not evaluated | Hypothermia                                                    | MOR (10 mg/kg) demonstrated analgesic properties from mechanical but not thermal pain. WIN55,212-2 did not produced significant analgesic effects against mechanical pain, but induced hypothermia in animals.                                                           | <a href="#">Maguma et al. (2010)</a>                                                   |
| Rat        | R-AM1241, S-AM1241, R,S-AM1241, MOR | mg/kg: 1 AM1241, 2 MOR                    | i.p. s.c. | Acute                    | Not evaluated | Analgesia                                                      | AM1241 & its enantiomers elicited antinociception to thermal induced, but not mechanical stimulation                                                                                                                                                                     | <a href="#">Rahn et al. (2010)</a>                                                     |
| Rat        | HU-210                              | 0.1 mg/kg                                 | i.p.      | Acute Chronic            | Not evaluated | Altered sexual behavior                                        | HU-210 stunted sexual behavior in male rats exposed to subchronic daily doses of HU-210                                                                                                                                                                                  | <a href="#">Riebe et al. (2010)</a>                                                    |
| Rat        | WIN55,212-2                         | 1 mg/kg                                   | i.p.      | Chronic                  | Not evaluated | Disease treatment                                              | WIN55,212-2 increased survival of brain cells in viral encephalitis-infected rats & supported oligodendrocyte survival                                                                                                                                                   | <a href="#">Solbrig et al. (2010)</a>                                                  |
| Rat        | HU-210                              | 25–100 µg/kg                              | i.p.      | Acute Subchronic Chronic | Not evaluated | Brain receptor (increased GABA <sub>A</sub> binding densities) | GABA <sub>A</sub> binding densities increased in hippocampus region of adult rats after chronic 100 µg/kg HU-210 exposure; in comparison, no significant changes in binding densities observed in adolescent rats under the same conditions                              | <a href="#">Verdurand et al. (2010)</a>                                                |
| Rat        | WIN55,212-2                         | 1 mg/kg                                   | i.p.      | Chronic                  | Not evaluated | Anxiolytic, Drug tolerance                                     | WIN55,212-2 reduced water maze anxiety in adolescent animal during the first 2 d of administration. No effect observed in adults. No effects in adolescents on subsequent days (3–5).                                                                                    | <a href="#">Acheson et al. (2011)</a>                                                  |
| Rat        | WIN55,212-2, AM251                  | 0.5 & 5 mg/kg                             | i.p.      | Acute Chronic            | Not evaluated | Cannabinoid Tetrad, Gastroparesis                              | WIN55,212-2 dose-dependently invoked cannabinoid tetrad. High dose (5 mg/kg) WIN55,212-2 delayed gastric emptying which progressed in chronic drug administrations. AM251 partially attenuated this effect.                                                              | <a href="#">Abalo et al. (2011)</a>                                                    |
| Mouse      | WIN55,212-2                         | 1 mg/kg                                   | s.c.      | Chronic                  | Not evaluated | Disease treatment                                              | WIN55,212 demonstrated anti-fibrosis properties in scleroderma bleomycin mouse model                                                                                                                                                                                     | <a href="#">Balistreri et al. (2011)</a>                                               |



Table 1 (Continued)

| Animal | Drug                                                          | Dose                                                                                                                         | Route | Exposure type | ED <sub>50</sub>                                                              | Measured effects                             | Results                                                                                                                                                                                                                                                                                                                                                               | Reference                   |
|--------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------|---------------|-------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Mouse  | CP55,940                                                      | 0.4 mg/kg                                                                                                                    | i.p.  | Chronic       | Not evaluated                                                                 | Hypothermia, Anxiolytic, Anxiogenic          | Tolerance to CP55,940 developed more rapidly in Nrg1 heterogeneous than in wild-type mice; CP55,940 lowered WT body temperature significantly than control group. Prepulse inhibition significantly increased by CP55,940 in Nrg1 mice compared to control & WT group. CP55,940-induced anxiety significantly increased in Nrg1 mouse than WT during open field test. | Boucher et al. (2011)       |
| Mouse  | JWH-018, JWH-018 4-hydroxyindole metabolite, THC (+ control)  | mg/kg: 3 JWH-018, 10 JWH-018 4-OHindole, 30 THC                                                                              | i.p.  | Acute         | Not evaluated                                                                 | Cannabinoid tetrad                           | JWH-018 & JWH-018 4-hydroxyindole produced cannabinoid tetrad greater than THC                                                                                                                                                                                                                                                                                        | Brents et al. (2011)        |
| Monkey | JWH-018, JWH-073, THC (+ control), RIM                        | THC discrimination (mg/kg): 0.032 JWH-018, 0.1 THC; THC-treated with RIM (mg/kg): 0.32–3.2 JWH-018, 3.2–32 JWH-073, 1–10 THC | i.v.  | Acute         | mg/kg: 0.032–0.061 THC, 0.0091–0.019 JWH-018, 0.036–0.094 JWH-073             | Drug discrimination                          | SC dose-dependently substituted for THC with potency order JWH-018 > THC > JWH-073; ED <sub>50</sub> dose increased in RIM-challenged animals                                                                                                                                                                                                                         | Ginsburg et al. (2011)      |
| Rat    | WIN55,212-2                                                   | 1 mg/kg                                                                                                                      | i.p.  | Acute         | Not evaluated                                                                 | Disease treatment                            | WIN55,212-2 showed cardioprotective properties in ischemia/reperfusion injury of diabetic rats                                                                                                                                                                                                                                                                        | Gonzalez et al. (2011)      |
| Rat    | WIN55,212-2, AM251                                            | mg/kg: 1 WIN55,212-2, 2 AM251                                                                                                | i.p.  | Acute         | Not evaluated                                                                 | Altered sleep pattern                        | WIN55,212-2 increased NREM sleep & lowered overall EEG spectral power; CB <sub>1</sub> inverse agonist AM251 did not prevent WIN55,212-2 sleep alterations                                                                                                                                                                                                            | Goonawardena et al. (2011)  |
| Mouse  | JWH-133                                                       | Not specified                                                                                                                | ?     | Acute         | Not evaluated                                                                 | Disease treatment                            | JWH-133, a CB <sub>2</sub> agonist, reduced formation of atherosclerotic lesion in ApoE-deficient mice, which was not prevented in ApoE-/CB2-deficient mice.                                                                                                                                                                                                          | Hoyer et al. (2011)         |
| Rats   | CP55,940                                                      | 0.4 mg/kg                                                                                                                    | i.p.  | Chronic       | Not evaluated                                                                 | Brain glucose metabolism                     | Acute 1 mg/kg cocaine exposure of SC-periadolescent-exposed female rats lowered brain glucose metabolism in septal nuclei, no observed effects in male rats                                                                                                                                                                                                           | Higuera-Matas et al. (2011) |
| Rat    | JWH-018 (AM678), AM5983, AM2233, WIN55,212-2, THC (+) control | mg/kg: 3THC, JWH-018, AM5983; 5.6–18 WIN55,212-2                                                                             | i.p.  | Acute         | mg/kg: 0.9–2 WIN55,212-2, 0.1–2 JWH-018 & AM5983, 0.3–0.8 AM2233, 1.0–1.3 THC | Drug discrimination                          | SC substituted for THC in the drug discrimination with potency order: AM5983 ≥ JWH-018 > AM2233 > WIN55,212-2 > THC                                                                                                                                                                                                                                                   | Järbe et al. (2011b)        |
| Rat    | WIN55,212-2                                                   | 1.2 mg/kg                                                                                                                    | i.p.  | Acute         | Not evaluated                                                                 | Behavioral deficits from adolescent exposure | Chronic pubertal SC exposure induced social recognition deficits & interaction deficits, but were attenuated with sub-acute 7 mg/kg quetiapine treatment                                                                                                                                                                                                              | Leweke and Schneider (2011) |
| Mouse  | WIN55,212-2                                                   | 2.5–10 mg/kg                                                                                                                 | i.p.  | Acute         | 4.6–10.0 mg/kg at 147.5 mg/kg dose + anti-convulsant drugs                    | Anti-convulsant/anti-epileptic               | WIN55,212-2 enhanced the anticonvulsant activity of 4 antiepileptic drugs; however, it also impaired motor coordination                                                                                                                                                                                                                                               | Luszczki et al. (2011)      |
| Mouse  | AM1710                                                        | 0.1–10 mg/kg                                                                                                                 | i.p.  | Acute         | Not evaluated                                                                 | Analgesia                                    | AM1710, has CB <sub>2</sub> receptor selectivity over CB <sub>1</sub> (K <sub>i</sub> = 17 ± 10 versus 282 ± 92 nM) dose-dependently produced analgesia in rats exposed to thermal stimulation without the effects of cannabinoid tetrad                                                                                                                              | Rahn et al. (2011)          |

|       |                                                                   |                                                                                                           |           |         |                                                                                                                                                                                                                          |                                  |                                                                                                                                                                                                                   |                                     |
|-------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Rat   | Cannabicyclohexanol, CP47,497, JWH-018, THC (+ control)           | 2.5 mg/kg                                                                                                 | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Catalepsy                        | SC increased EEG in the first 3 h between 5 and 6.0 Hz; THC decreased EEG power between 7.0 and 20.0 Hz at 1 h; SC reduced locomotor activity longer than THC with potency order: CCH > CP47,497 > JWH-018 > THC. | Uchiyama et al. (2011)              |
| Rat   | HU-210, CP55,940                                                  | 0.1 mg/kg                                                                                                 | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Locomotion, Catalepsy            | CP 55,940 & HU-210 produced decreased locomotion & catalepsy                                                                                                                                                      | Bosier et al. (2012)                |
| Rat   | CP55,940, THC, MOR, RIM                                           | 20–40 mg/kg THC, 0.4–1.6 mg/kg CP55,940                                                                   | i.p.      | Acute   | Male (mg/kg): 0.06–>200 CP55,940, 0.09 > 200 CP55,940 + 1.0 mg/kg RIM, 0.31–103.28 CP55,940 + 10 mg/kg RIM; female, mg/kg: 0.04–0.49 CP 55,940, 0.18–1.92 CP55,940 + 1.0 mg/kg RIM, 0.35–91.4 CP55,940 + 10.0 rimonabant | Analgesia                        | CP55,940 & THC antinociceptive effects were greater in female than male animals with or without RIM                                                                                                               | Craft et al. (2012)                 |
| Mouse | THC, CP 55,940, WIN55,212-2, HU-210, JW-133, methanandamide, ACEA | mg/kg: 1–6 THC, JWH-133, & methanandamide, 10–50 CP 55,940, 0.5–3 WIN55,212-2, 0.01–0.1 HU-210            | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Appetite increase/loss           | THC, CP55,940, & WIN55,212-2 increased milk consumption, but not with HU-210, JWH-133, ACAE & methanandamide                                                                                                      | Grey et al. (2012)                  |
| Mouse | JWH-018, THC                                                      | 3 mg/kg                                                                                                   | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Appetite loss                    | JWH-018 exposed mice developed food aversion, but lesser intensity & short-lived in THC pre-treated mice                                                                                                          | Hyatt et al. (2012)                 |
| Rats  | AM2389, AM5983 (+ control), THC (+ control)                       | Temperature (mg/kg): 0.1, 0.3 AM2389, 30 THC; drug discrimination (mg/kg): 0.18 & 0.56 AM5983 (+ control) | i.p. p.o. | Acute   | 0.0038–0.0094 mg/kg AM2389 at 1 h versus 0.18 mg/kg AM5983; 0.0053–0.0090 mg/kg AM2389 versus 0.56 mg/kg                                                                                                                 | Hypothermia, Drug discrimination | AM2389 produced hypothermia & fully substituted for AM5983 & THC with potency order AM2389 > AM5983 > THC                                                                                                         | Jarbe et al. (2012)                 |
| Rats  | CP55,940, THC                                                     | mg/kg: 0.0032–0.32 CP55,940, 0.1–10 THC                                                                   | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Analgesia                        | THC & CP55,940 dose-dependently produced analgesia in acid-induced pain stimulation than acid-induced depressed operant behavior                                                                                  | Kwilas and Negus (2012)             |
| Rats  | HU-210                                                            | 50 µg/kg                                                                                                  | i.p.      | Chronic | Not evaluated                                                                                                                                                                                                            | Reproductive/sexual maturity     | Animal exposed to HU-210 from beginning of sexual maturity had lower body & kidney weight than control group; sperm counts also were reduced by 46% after 7 wk of HU-210 exposure                                 | Lewis et al. (2012)                 |
| Mouse | WIN55,212-2, pregabalin                                           | 1.28–15 mg/kg WIN55,212-2, 6.25–75 mg/kg pregabalin                                                       | i.p.      | Acute   | 10.5 mg/kg (ED30); 5.7 mg/kg (ED30 for 1:1 pregabalin: WIN 55,212-2)                                                                                                                                                     | Analgesia                        | WIN55,212-2 co-administered with pregabalin dose-dependently produced additive antinociceptive effect in animal thermal pain model                                                                                | Luszczki and Florek-Luszczki (2012) |
| Rat   | HU-210                                                            | 0.1 mg/kg                                                                                                 | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Glucose metabolism               | Acute high dose HU-210 administration increased brain glucose metabolism by diminished after 24 h post exposure                                                                                                   | Nguyen et al. (2012)                |
| Mouse | JWH-018; THC (+ control), RIM                                     | 10–50 mg plant material containing 5.8% JWH-018; 200 mg cannabis plant (7.4%), 3 mg/kg RIM                | inhale    | Acute   | Not evaluated                                                                                                                                                                                                            | Cannabinoid tetrad               | (+) cannabinoid tetrad (dose-dependent) & ptosis, hyper-reflexive response, Straub-tail; effects blocked by RIM                                                                                                   | Wielbelhaus et al. (2012)           |
| Mouse | JWH-018, 1-pentyl-3-phenylacetylindoles                           | see ED <sub>50</sub>                                                                                      | i.v.      | Acute   | µmol/kg: 0.1–1.8 JWH-018, 1.3–13 JWH-167, 0.1–6.0 JWH-203, 0.8–2.0 JWH-204, 13–19 JWH-205, 0.9–6 JWH-251, 2.8–38 JWH-208, 1.5–3 JWH-237, 1.1–2.9 JWH-306, 0.9–2.6 THC                                                    | Cannabinoid tetrad               | (+) cannabinoid tetrad                                                                                                                                                                                            | Wiley et al. (2012)                 |
| Mouse | AB-001, AB-002, SDB-001                                           | mg/kg: 1–10 THC, 0.3–3 JWH-018, 1–10 SDB-001, 0.3–30 AB-001, 3 & 30 AB-002                                | i.p.      | Acute   | Not evaluated                                                                                                                                                                                                            | Hypothermia, Bradycardia         | SDB-001 dose-dependently induced & prolonged hypothermia > THC & JWH-018. SDB-001, JWH-018, & THC also reduced heart-rate, but not with AB-001 or AB-002                                                          | Banister et al. (2013)              |

Table 1 (Continued)

| Animal | Drug                                                        | Dose                                                                                                                                                                                                                  | Route       | Exposure type | ED <sub>50</sub>                                                                                                     | Measured effects                                     | Results                                                                                                                                                                                                                     | Reference                                         |
|--------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Mouse  | JWH-081, RIM                                                | mg/kg: 0.625, 1.25 JWH-081, 3.0 RIM                                                                                                                                                                                   | i.p.        | Acute         | Not evaluated                                                                                                        | Object recognition & spatial memory impairment       | JWH-081 produced dose-dependent impairment in the object recognition task, spontaneous alteration & spatial recognition memory; object-recognition deficit attenuated by RIM pre-treatment                                  | <a href="#">Basavarajappa and Subbanna (2013)</a> |
| Mouse  | JWH-018, JWH-073, THC (+ control)                           | mg/kg: 1–10 SC, 10 THC                                                                                                                                                                                                | i.p.        | Acute         | mg/kg: 0.6 ± 0.2 JWH-018, 2.0 ± 0.6 JWH-073                                                                          | Cannabinoid tetrad, <sup>a</sup> Drug discrimination | SC substituted for THC in a drug discrimination study producing synergistic effects when administered concomitantly in 1:1 & 1:3 ratios. SC also produced cannabinoid tetrad, but effects were not synergistic or additive. | <a href="#">Brents et al. (2013)</a>              |
| Mouse  | AM4054, AM7418, WIN55,212-2, THC, RIM                       | mg/kg: 0.01–1 AM4054, 0.03–1.0 AM7418, 1–10 THC, 0.3–10 WIN55,212-2, 1–10 RIM                                                                                                                                         | s.c.        | Chronic       | 0.06, 0.04 mg/kg AM4054                                                                                              | Analgesia, Diuresis                                  | SC dose-dependently elicited antinociception & produced dose-related diuresis in a 6 h period after a single administration. RIM pre-treatment did not alter urine output.                                                  | <a href="#">Chopda et al. (2013)</a>              |
| Monkey | AM411, AM4054, WIN55,212-2, THC (+ control), methanandamide | Acute (mg/kg): 0.32–1.0 AM411, 0.0032–0.1 AM4054, 0.32–1.0 WIN 55,212-2, 0.32–32 THC, 1.0–10 methanandamide; Chronic (mg/kg): 0.32–100 AM411, 0.1–3.2 AM4054, 1.0–32 WIN 55,212-2, 3.2–100 THC, 10–320 methanandamide | i.m.        | Acute Chronic | mg/kg: 0.08–5.24 AM411, 0.01–0.07 AM4054, 0.1–1.2 WIN55,212-2, 0.33–2.11 THC, methanandamide (no significant effect) | Drug tolerance                                       | SC dose-dependently decreased response rate to stimulus shock, animal developed tolerance after chronic exposure; potency order: AM4054 > AM411 ~ WIN55,212-2 > THC > methanandamide                                        | <a href="#">Desai et al. (2013)</a>               |
| Rat    | CP55,940                                                    | 0.05 mg/kg; challenged with 0.35 mg/kg DOI 30 min prior to sacrifice                                                                                                                                                  | i.p.        | Chronic       | Not evaluated                                                                                                        | Serotonin-mediated interaction                       | DOI-challenged rats had increased (p < 0.05) 5-HT <sub>2A</sub> receptor-mediated prolactin & corticosterone plasma levels                                                                                                  | <a href="#">Franklin et al. (2013)</a>            |
| Rat    | WIN55,212-2                                                 | 0.5 µg/kg                                                                                                                                                                                                             | i.h.        | Acute         | Not evaluated                                                                                                        | Catalepsy, Anti-epileptic                            | WIN55,212-2 produced sedation & attenuated NMDA-induced epileptic currents                                                                                                                                                  | <a href="#">Friedman and Rudenko (2013)</a>       |
| Rat    | CP55,940                                                    | 0.1–0.3 mg/kg                                                                                                                                                                                                         | i.p.        | Chronic       | Not evaluated                                                                                                        | Disease treatment                                    | CP55,940 decreased fimbria (hippocampus) fractional anisotropy value                                                                                                                                                        | <a href="#">Humbert-Claude et al. (2013)</a>      |
| Rat    | HU-210                                                      | 100 µg/kg                                                                                                                                                                                                             | i.p.        | Chronic       | Not evaluated                                                                                                        | Disease treatment                                    | Ameliorated vision loss for P23H rat model of retinitis pigmentosa                                                                                                                                                          | <a href="#">Lax et al. (2013)</a>                 |
| Mouse  | JWH-018, WIN55,212-2                                        | mg/kg: 0.03–0.3 JWH-018, 2 WIN55,212-2                                                                                                                                                                                | i.p.        | Acute         | Not evaluated                                                                                                        | Cannabinoid tetrad                                   | (+) cannabinoid tetrad; reduced sedative effects of JWH-018 & WIN55,212-2 in adult mouse, prenatally exposed to corticosterone                                                                                              | <a href="#">Macri et al. (2013)</a>               |
| Monkey | WIN55-212, CP55,940, Heroin, RIM                            | mg/kg: 0.01–0.32 CP55,940, 0.1–1.0 WIN55,212-2; 3.2 MOR, & 0.32–10 µg/kg/infusion heroin, 1 RIM                                                                                                                       | i.v. & s.c. | Acute         | Not evaluated                                                                                                        | Analgesia, Drug discrimination/cross tolerance       | SC & MOR dose-dependently elicited analgesic effects; SC dose-dependently reduced frequency of heroin self-administration; effects attenuated by RIM pre-treatment                                                          | <a href="#">Maguire et al. (2013)</a>             |
| Rat    | AM4054                                                      | 0.01–0.16 mg/kg                                                                                                                                                                                                       | i.p.        | Acute         | Not evaluated                                                                                                        | Impairment                                           | AM4054 dose-dependent impairment in animal for both two-choice operant experiments                                                                                                                                          | <a href="#">Miller et al. (2013)</a>              |

|        |                                             |                                                       |      |                                |                                                                                                                                                                                                 |                                         |                                                                                                                                                                                                                |                              |
|--------|---------------------------------------------|-------------------------------------------------------|------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Rat    | CP55,940                                    | 0.15–0.3 mg/kg                                        | i.p. | Chronic                        | Not evaluated                                                                                                                                                                                   | Short-term & spatial memory deficits    | Adolescent rats exposed to CP55,940 developed short-term & spatial memory impairment when tested as adults. In contrast, no long-term deleterious effects were observed in CP55,940 exposed adult animals.     | Renard et al. (2013)         |
| Monkey | JWH-018, JWH-073, THC (+ control), CP55,940 | 0.01–1.0 mg/kg                                        | i.v. | Acute<br>Subchronic<br>Chronic | Subchronic (mg/kg): 0.046–0.13 THC, 0.005–0.006 CP55,940, 0.024–0.042 JWH-018, 0.048–0.077 JWH-073, Chronic (mg/kg): 0.42–0.39 THC, 0.004–0.13 CP55,940, 0.014–0.074 JWH-018, 0.08–0.32 JWH-073 | Drug discrimination                     | Synthetic cannabinoids substituted for THC with developed cross-tolerance & decreased sensitivity after 14d. JWH-018, CP55,940 > JWH-073 > THC                                                                 | Rodriguez and McMahon (2013) |
| Mouse  | UR-144, XLR-11, THC, RIM                    | mg/kg: 5.6 SC & THC, 3 RIM                            | i.p. | Acute                          | $\mu$ mol/kg: 0.6–3.3 XLR-11, 0.6–2.6 UR-144, 3–15 THC                                                                                                                                          | Cannabinoid tetrad, Drug discrimination | (+) cannabinoid tetrad, XLR11 = UR-144 > THC; effects blocked by RIM; drug discrimination study (animal trained to discriminate THC), XLR-11 substituted for THC at 3.5 $\mu$ mol/kg & 7.4 $\mu$ mol/kg UR-144 | Wiley et al. (2013a)         |
| Rat    | AM5983, WIN55,212-2, THC (+ control)        | mg/kg: 0.1–.56 AM5983, 0.1–0.3 THC, 0.1–1 WIN55,212-2 | p.o. | Acute<br>Subchronic            | mg/kg: 0.04–0.13 AM5983, 0.15–0.84 THC, 0.13–0.32 WIN55,212-2                                                                                                                                   | Drug discrimination                     | AM5983 dose-dependently substituted for THC; potency order: AM5983 > WIN > THC                                                                                                                                 | Järbe et al. (2014)          |

**Abbreviations:** 8-OH-DPAT [7-(Dipropylamino)-5,6,7,8-tetrahydronaphthalen-1-ol], AB-001 [1-Pentyl-3-(1-adamantoyl)indole], AB-002 [2-(Adamantan-1-yl)-1-(1-pentyl-1H-indol-3-yl)ethanone], ACEA (arachidonyl-2'-chloroethylamide), AM251 [1-(2,4-Dichlorophenyl)-5-(4-iodophenyl)-4-methyl-N-(1-piperidyl)pyrazole-3-carboxamide], AM404 [(all-Z)-(4-Hydroxyphenyl)-5,8,11,14-eicosatetraenamide], AM411 [(6aR,10aR)-3-(1-Adamantyl)-6,6,9-trimethyl-6a,7,10,10a-tetra-hydro-benzo[c]-chromen-1-ol], AM1387 [1'-(2,4-Dichlorophenyl)-4-(hydroxymethyl)-N-(piperidinyl)-5-arylphenyl-1H-pyrazole-3-carboxamide], AM1710 [3-1(1,1-Dimethylheptyl)-1-hydroxy-9-methoxy-6H-benzo[c]-chromene-6-one], AM2233 [(2-Iodophenyl)-1-[(1-methyl-2-piperidinyl)methyl]-1H-indol-3-yl]methanone], AM2389 [9 $\beta$ -Hydroxy-3-(1-hexyl-cyclobut-1-yl)-hexahydrocannabinol], AM4054 [9b-(Hydroxymethyl)-3-(1-adamantyl)-hexahydrocannabinol], AM4113 [5-(4-Alkylphenyl)-1-(2,4-dichlorophenyl)-4-methyl-N-(piperidin-1-yl)-1H-pyrazole-3-carboxamide], AM5983 [1-[(1-methylpiperidin-2-yl)methyl]-1H-indol-3-yl(naphthalen-1-yl)methanone], AMP (amphetamine), APDC [(2R,4R)-4-Aminopyrrolidine-2,4-dicarboxylate], CP55,940 [2-[(1R,2R,5R)-5-Hydroxy-2-(3-hydroxy-propyl)-cyclohexyl]-5-(2-methyl-octan-2-yl)-phenol], COC (cocaine), DOI [1-(4-Iodo-2,5-dimethoxyphenyl)-2-propanamine], EEG (electroencephalogram), ETOH (ethanol), GABA (gamma-amino butyric acid), GAD (glutamic acid decarboxylase), i.c.v. (intracerebroventricular), HU-210[(6aR)-trans-3-(1,1-Dimethylheptyl)-6a,7,10,10a-tetrahydro-1-hydroxy-6,6-dimethyl-6Hdibenzo[b,d]pyran-9-methanol], HU-320 (3S,4S)-3-[2,6-Dihydroxy-4-(2-methyloctan-2-yl)-phenyl]-4-(prop-1-en-2-yl)cyclohex-1-ene-1-carboxylic acid], i.h. (intrahippocampal), i.m. (intramuscular), i.p. (intraperitoneal), i.t. (intrathecal), i.v. (intravenous), JWH-015 [Naphthalen-1-yl(2-methyl-1-propylindol-3-yl)methanone], JWH-018 [Naphthalen-1-yl(1-pentylindol-3-yl)methanone], JWH-073 [1-Butylindol-3-yl]naphthalen-1-ylmethanone, JWH-081[N-Pentyl-3-[1-(4-methoxy)-naphtho[1,2-b]indole], JWH-133 [(6aR,10aR)-3-(1,1-Dimethylbutyl)-6a,7,10,10a-tetrahydro-6,6,9-trimethyl-6H-dibenzo[b,d]pyran], JWH-167 [1-(1-Pentyl-1H-indol-3-yl)-2-phenyl-ethanone], JWH-203 [1-Pentyl-3-(2-chlorophenylacetyl)indole], JWH-204 [2-(2-Chlorophenyl)-1-(2-methyl-1-pentylindol-3-yl)ethanone], JWH-205 [1-(2-Methyl-1-pentyl-1H-indol-3-yl)-2-phenylethanone], JWH-208 [2-(4-Methylphenyl)-1-(1-pentylindol-3-yl)ethanone], JWH-251 [2-(2-Methylphenyl)-1-(1-pentyl-1H-indol-3-yl)-ethanone], JWH-237 [1-Pentyl-3-(3-chlorophenylacetyl)indole], JWH 306 [2-(2-Methoxyphenyl)-1-(1-pentyl-1H-indol-3-yl)-ethanone], L-AP4 [1-(+)-2-Amino-4-phosphonobutyric acid], m.d. (microdialysis), MOR (morphine), NMDA (N-Methyl-D-aspartate), RIM (rimonabant), s.c. (subcutaneous), SDB-001 [N-(Adamantan-1-yl)-1-pentyl-1H-indole-3-carboxamide], SR144528 [N-[(1S)-Endo-1,3,3-trimethylbicyclo [2.2.1]heptan-2-yl]-5-(4-chloro-3-methylphenyl)-1-[(4-methylphenyl)methyl]-1H-pyrazole-3-carboxamide], TMEV (Theiler's encephalomyelitis virus), TMJ (temporomandibular joint), UR-144 [(1-(5-Chloropentyl)-1H-indol-3-yl)(2,2,3,3-tetramethyl-cyclopropyl)-methanone], XLR11 [(1-(5-Fluoropentyl)-1H-indol-3-yl)(2,2,3,3-tetramethylcyclopropyl)-methanone], WIN55,212-2 (R)-(+)-[2,3-Dihydro-5-methyl-3-(4-morpholinylmethyl)pyrrolol(1,2,3-de)-1,4-benzoxazin-6-yl]-1-naphthalen-ylmethanone], WT (wild-type).

<sup>a</sup> Cannabinoid tetrad: hypomotility, catalepsy, hypothermia, analgesia.

i.p.) chronically-treated adolescent rats than adults (Schneider and Koch, 2005).

Chronic exposure to HU-210 (25, 50, or 100 µg/kg i.p.), a CB<sub>1</sub>/CB<sub>2</sub> agonist 100 times more potent than THC *in vivo*, reduced body weight in adult rodents during the first 4 days, with slow weight gain thereafter, although still lower than controls after 14 days (Dalton et al., 2009). Weight loss was associated with dose-dependent CB<sub>1</sub> receptor down-regulation continuing throughout chronic exposure. In contrast, adolescent rats gained weight over the entire study, but also showed some CB<sub>1</sub> receptor down-regulation over 14 days. Single 100 µg/kg i.p. HU-210 doses significantly decreased CB<sub>1</sub> binding in the caudate putamen and hippocampus of adolescent rats and increased brain glucose metabolism in young adult rats, with metabolism returning to normal the following day (Nguyen et al., 2012). These studies suggest that SC's chronic effects are influenced by age, dose, and exposure duration, and that drug tolerance can develop over time.

Animals exposed to 100 µg/kg i.p. HU-210 for 12 days manifested increased anxiety, correlating with higher plasma cortisone concentrations compared to controls (Hill and Gorzalka, 2006). Repeated 3.0 mg/kg i.p. WIN55,212-2 induced anxiogenic-like behaviors in rats, associated with increased catecholamine expression in locus coeruleus, which is involved in norepinephrine synthesis (Page et al., 2007). CP55,940, a nonselective cannabinoid agonist, administered acutely (10–50 µg/kg i.p.) to an anxiety-prone rat strain elicited anxiogenesis (Arnold et al., 2010; Boucher et al., 2011), which diminished as tolerance developed (Boucher et al., 2011). Repeated 7-day exposure to 50 µg/kg i.p. CP55,940 was associated with 5-HT<sub>2A</sub> receptor upregulation in rat prefrontal cortex (Franklin and Carrasco, 2013) and hypothalamic paraventricular nucleus (Franklin et al., 2013), as well as enhanced anxiety-like behaviors. These observations suggest a link between SC and anxiety that is mediated by brain serotonergic activity, which is influenced by baseline anxiety level.

**3.2.2. Drug discrimination and drug re-instatement studies.** In rat discrimination studies, AM5983 [(1-[(1-Methylpiperidin-2-yl)methyl]-1H-indol-3-yl)(naphthalen-1-yl)methanone] and AM2389 [9β-Hydroxy-3-(1-hexyl-cyclobut-1-yl)-hexahydrocannabinol] with ED<sub>50</sub> 0.06 and 0.03 mg/kg, respectively, potency was 4–105 times higher than THC's (0.26 mg/kg) (Järbe et al., 2011a). In non-human primates, JWH-073 and JWH-018 induced THC-like subjective effects at concentrations as low as 0.032 mg/kg (JWH-018), compared to 0.1 mg/kg THC with a shorter duration of action (1.5–2.3 h *versus* 3 h for THC). Rimnabant (0.32–1 mg/kg i.v.), a CB<sub>1</sub> receptor antagonist/inverse agonist, dose-dependently attenuated SC discriminative stimulus effects (Ginsburg et al., 2011), suggesting mediation through the CB<sub>1</sub> receptor.

Rimnabant-elicited withdrawal symptoms (head shakes, increased heart rate) in monkeys previously exposed to THC were dose-dependently attenuated by 0.032–0.32 mg/kg CP55,940 or 1–10 mg/kg WIN55,212-2 i.v. administration. CP55,940 (0.4 mg/kg i.p.) and WIN55,212-2 (2.0–8.0 mg/kg i.p.) exposure of adolescent rats facilitated desensitization to other illicit drugs (Bisicaia et al., 2008; Pistis et al., 2004). HU-210 (4–100 µg/kg i.p.) dose-dependently reinstated previously extinguished cocaine self-administration (animal relapse model) in rats, which was blocked by 1 mg/kg rimnabant (De Vries et al., 2001).

**3.2.3. In utero exposure.** In contrast to adolescent exposure, there appear to be little or no adverse effects from SC exposure *in utero*. Daily oral 1, 5 and 25 µg/kg HU-210 exposure to pregnant rats had no significant effect on gestational progression or post-natal food and water intake when compared to controls (del Arco et al., 2000). Offspring exposed to HU-210 *in utero* showed no significant difference from the non-exposed control group in average birth

weight, body length, and lymphocyte immune function; however, 1 µg/kg HU-210-exposed rats had a 17% increase in spleen size. WIN55,212-2 (0.5 mg/kg s.c.) daily administration to rats during gestation had no effect on gestational progression, birthweight, or brain development into adulthood; however, glutamic acid decarboxylase (GAD-65/67) and γ-aminobutyric acid (GABA) expression in the cerebellar cortex was significantly higher than in controls (Benagiano et al., 2007).

In summary, animal studies indicate that SC acute or chronic exposure during adolescence or adulthood, but not *in utero*, induces dose-dependent physiological and behavioral effects and brain CB<sub>1</sub> receptor downregulation. Chronic SC exposure during adolescence can have detrimental behavioral effects as an adult.

### 3.3. Human pharmacodynamics

The American Association of Poison Control Call Centers received 2906 SC-related calls in 2010, 6968 in 2011, and 5230 in 2012, and 2639 in 2013 (Supplementary Material Figure 1)<sup>2</sup> (American Association of Poison Control Centers, 2013; Bronstein et al., 2012). There were 11,406 SC-related emergency department (ED) visits in 2010, increasing to 28,531 in 2012 (Substance Abuse and Mental Health Services Administration, 2012). In 2010, Texas poison control centers received 464 reports of SC-related exposures: 73.9% men and 25% women, ranging in age from 12 to 67 years (Forrester et al., 2012). Adverse clinical effects were categorized as neurological (61.9%), cardiovascular (43.5%), gastrointestinal (21.1%), respiratory (8.0%), ocular (5.0%), dermal (2.6%), renal (0.9%), hematological (0.4%) and miscellaneous, e.g., acidosis, hyperglycemia, diaphoresis (25.9%). No SC-related fatality was reported, although 59.9% of patients manifested “moderate or major” toxicity symptoms. Treatments included i.v. fluids (38.8%), benzodiazepines (18.5%), oxygen (8.0%), and anti-emetics (6.0%).

We identified 63 peer-reviewed articles and 4 meeting abstracts, which included case reports/series, retrospective toxicological data reviews, driving under the influence of drugs (DUID) reports, criminal/forensic cases, and self- and controlled drug administration studies. The majority of case reports were ED visits; other toxicity data came from poison control center calls, inpatient psychiatric cases, and law enforcement drug recognition examiner (DRE) evaluations.

**3.3.1. Acute intoxication.** We identified 51 articles from 5 continents reporting >200 acute intoxication cases (summarized in Table 2). ED patients presenting with SC intoxication were ages 13–59 years (mean 22, median 20). Acute SC-intoxication psychoactive symptoms included agitation or irritability, restlessness, anxiety, confusion, short-term memory and cognitive impairment, and psychosis. Physical signs included dilated pupils, reddened conjunctivae, nausea and vomiting, slurred speech, shortness of breath, hypertension, tachycardia (up to 180 bpm), chest pain, muscle twitches, and sweating or skin pallor. Physical examination (except as noted above), clinical laboratory tests, and electrocardiogram (ECG) were generally normal, except for some patients with mild leucocytosis (WBC count 13,000–14,000) or hypokalemia (<3.5 meq/L). Hyperglycemia also was observed. Urine toxicological screens were often negative for illicit drugs. Only 26 acute intoxication articles reported SC detection in serum, blood, oral fluid or urine, including ADB-PINACA [N-(1-Amino-3,3-dimethyl-1-oxobutan-2-yl)-1-pentyl-1H-indazole-3-carboxamide], AM2201 [N-(5-Fluoropentyl)-3-(1-naphthoyl)lindole], CP47,497 C8 homolog

<sup>2</sup> Supplementary material can be found by accessing the online version of this paper. Please see Appendix A.



[5-(1,1-Dimethyloctyl)-2-[(1R,3S)-3-hydroxycyclohexyl]-phenol], JWH-018, JWH-019 [(1-Hexylindol-3-yl)naphthalene-1-ylmethanone], JWH-073, JWH-081 [4-Methoxy-1-naphthalen-1-yl-(1-pentylindol-3-yl)methanone], JWH-122 [1-Pentyl-3-(1-(4-methylnaphthoyl))indole], JWH-210 [4-Ethyl-naphthalen-1-yl-(1-pentylindol-3-yl)methanone], JWH-250 [1-Pentyl-3-(2-methoxyphenylacetyl)indole], JWH-307 [(5-(2-Fluorophenyl)-1-pentylpyrrol-3-yl)-1-(naphthalenylmethanone)], MAM2201 [1-(5-Fluoropentyl)-3-(4-methyl-1-naphthoyl)indole], RCS-4 [4-Methoxyphenyl-(1-allyl-1H-indol-3-yl)methanone], XLR-11 [(1-(5-Fluoropentyl)-1H-indol-3-yl)(2,2,3,3-tetramethylcyclopropyl)methanone], and UR-144 [(1-Pentylindol-3-yl)-(2,2,3,3-tetramethylcyclopropyl)methanone]. Metabolites were identified in urine except following ADB-PINACA, JWH-250, JWH-307, and RCS-4 intake. Patients generally received supportive care, benzodiazepines, and i.v. saline, except in cases with serious complications such as seizures that often also required intubation. Some patients reported effect onset within minutes of smoking and intoxication for 2–5 h, with most recovering in <24 h.

Serious medical complications developed after subacute exposures (>24 h after intake), including myocardial infarction, ischemic strokes, seizures, and acute kidney injury (AKI). Among patients with AKI, XLR11 and/or UR-144 (parent and metabolites) were detected in urine, blood, or serum and implicated as causing injury. Some patients with AKI required hemodialysis and corticosteroid treatment, while others recovered within 3 days of hospital admission.

**3.3.2. Controlled administration studies.** We identified five peer-reviewed articles and one conference proceeding abstract describing SC controlled administration (Table 3). Two German investigators smoked a shared cigarette containing 0.3 g “Spice Diamond,” a mixture of a dimethyl CP47,497 [2-[(1R,3S)-3-Hydroxycyclohexyl]-5-(2-methyloctan-2-yl)phenol] homologue, the trans-diastereomer of the CP47,497 homologue, and small amounts of JWH-018 (Auwärter et al., 2009). Acute effects, noticeable 10 min after smoking, included altered mood and perception, tachycardia, dry mouth, and reddened conjunctivae. There was no impairment in objective psychomotor tests, although the subjects felt they were moderately impaired. Objective effects resolved over 6 h; however, the investigators reported noticeable “minor after-effects” the next day. Two other investigator-involved studies included oral AB-001 [(1-Pentyl-1H-indol-3-yl)adamantylmethanone] and oral and smoked AM694 [1-(5-fluoropentyl)-3-(2-iodobenzoyl)indole]. AB-001 was ingested by 1 male (13 mg) and 1 female (26 mg) (Grigoryev et al., 2012b). AM694 (10 mg) was first ingested by one male investigator, who later smoked a drug-laced cigarette soaked in acetone containing 1 g/L AM694 (Grigoryev et al., 2012a). In these studies, investigators reported no physiological effects. After 5 mg oral AM2201, no psychological or physiological effects were reported, with serum concentrations of AM2201 (0.04 µg/L), 6-hydroxyindole AM2201 (0.2 µg/L), JWH-018 pentanoic acid (0.7 µg/L), and JWH-018 5-hydroxypentyl (0.3 µg/L), 1.5 h after ingestion (Hutter et al., 2013). AM2201 was not present after 28 h, while AM2201 6-hydroxyindole and JWH-018 metabolites were present up to 28 and 57 h, respectively.

In a German study, a 33-year old woman and a 47-year old man smoked 100 and 150 mg, respectively, of a 2.9% JWH-018 cigarette, yielding 50 µg/kg body weight JWH-018. Immediate effects included sedation, dry mouth, feeling sick, subjective thought disruption, burning eyes, hot flushes, and tachycardia (Teske et al., 2010). No significant change in blood pressure or pupil reaction was noted. After acute effects resolved, subjects felt tired and exhausted for 6–12 h after smoking. JWH-018 serum concentrations after 5 min were 8.1 (Subject 1) and 10.2 (Subject 2) µg/L.

Concentrations rapidly declined at 1 h after intake and were undetectable at 24 h.

An unpublished controlled drug administration study was conducted at the University of Central Missouri (Warrensburg, MO) in collaboration with NMS Laboratories (Willow Grove, PA; Logan et al., 2011). Six healthy research volunteers took up to three inhalations from a water pipe containing 0.3 g SC-laced herbal product. Two subjects each smoked either K2 Standard (9 mg/g JWH-018 and 9 mg/g JWH-073), K2 Citron (10 mg/g of each), or K2 Summit (11 mg/g JWH-018 and 9 mg/g JWH-073). There was no placebo condition and it is not clear whether raters were blinded to study conditions. Impairment was assessed by a DRE. Acute subjective effects began at 2–3 min and peaked 5–10 min after smoking and included changes in mood and perception (5 subjects), self-assessed impairment (5), mild anxiety (4), impaired sense of time (4), loss of concentration (4), sedation (3), and paranoia (2). No subject experienced hallucinations. Physical effects included tachycardia (6 subjects), reddened conjunctivae (6), dry mouth (4), and burning eyes (1). Increased systolic blood pressure, body sway and other signs of impaired balance and motor coordination, leg and body tremor, and lack of eye convergence occurred in some subjects. Pupil size and reactivity, muscle tone, and body temperature remained normal; no horizontal or vertical gaze nystagmus was observed. Acute effects resolved within a few hours, but three subjects experienced fatigue and exhaustion (hangover-like effect) for 6–12 h after smoking. JWH-018 and JWH-073 were quantified in blood at 4–5 µg/L after 1 h, while metabolites were detected in urine (concentrations not specified).

**3.3.3. Driving impairment.** We identified 3 peer-reviewed articles documenting DUID with SC. The first involved 8 drivers in Germany stopped by police for suspected DUID (Musshoff et al., 2013). Somnolence and retarded movements were the most commonly observed symptoms. SC were identified in serum (collected within 2 h) in all suspected cases, with concentrations ranging from 0.33 µg/L to 28 µg/L. Six of 8 suspects' serum contained multiple SC. One motorcyclist's serum contained 7 SC: JWH-122 (28 µg/L), JWH-210 (2.5 µg/L), and JWH-018 (1.9 µg/L). AM2201, JWH-307, MAM2201, and UR-144 also were detected in low concentrations (<0.1 µg/L). The examining physician reported “no abnormalities” 1.5 h after arrest.

In the US, NMS Laboratories confirmed SC in blood from 12 suspects receiving DRE evaluations between 2010 and 2011 (Yeakel and Logan, 2013). AM2201, JWH-018, JWH-081, JWH-122, JWH-210, JWH-250 blood concentrations were 0.24–9.9 µg/L. The predominant SC was AM2201 (7 samples, 0.4–4.0 µg/L). All suspects failed their DRE examination, with general poor motor coordination, but the majority were “cooperative and relaxed.” About 25% were involved in motor vehicle accidents; none were positive for alcohol. These symptoms were different from ED reports of agitation, aggression, and panic attacks associated with SC intoxication.

In a Norwegian study conducted between November, 2011 and April, 2012, 2.2% of 726 blood samples collected from DUID suspects confirmed positive for SC (Tuv et al., 2014). Each sample quantified for only one SC, with AM2201 (0.07–1.33 µg/L) and JWH-018 (0.08–0.46 µg/L) the most prevalent ( $n = 5$  each). All samples also were positive for other psychoactive drugs: THC (12), benzodiazepines (12), methamphetamine and amphetamine (4), methamphetamine only (3), LSD (2), ketamine (1), methadone (1), methylphenidate (1), and alcohol (1). The majority (11) were impaired and/or involved in traffic accidents.

**3.3.4. Subacute psychiatric effects.** Five reports from psychiatric inpatient units and two ED visits suggested that some SC smokers experienced longer-lasting psychiatric effects. Fourteen



**Table 2**  
Synthetic cannabinoid (SC) acute and sub-acute intoxication documented from case reports/series and retrospective case reviews.

| Country | N  | Sex    | Age    | SC Brand                                               | Dose (Route)       | Symptoms                                                                                                                                                                                                                                                                          | Time onset           | Duration | Laboratory results (matrix)                                           | Treatment                                           | Reference                      |
|---------|----|--------|--------|--------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------|-----------------------------------------------------------------------|-----------------------------------------------------|--------------------------------|
| USA     | 11 | M      | 13–27  | Spice                                                  | UNK                | Tachycardia (6), anticholinergic toxidrome (4), agitation/irritability (4), tremor (4), confusion(3), pallor (2), mydriasis(2), hypertension (2)                                                                                                                                  | UNK                  | UNK      | (–)UDS(3); SC not tested for all                                      | BDP(3), supportive care (10)                        | Banerji et al. (2010)          |
| USA     | 1  | M      | 18     | K2 Summit                                              | 30 gm (sm)         | Tremors, blurred vision, nausea, vomiting, incoherent speech                                                                                                                                                                                                                      | 30 min               | 4.5 h    | (–)UDS, 0.5 µg/L JWH-018(serum)                                       | Anti-emetics, IVF                                   | Canning et al. (2010)          |
| Germany | 1  | M      | 21     | UNK                                                    | 40 mg (sm)         | Blurred vision, unsteady gait, excessive sweating, heart palpitations, anxiety                                                                                                                                                                                                    | “within min”         | <24 h    | (–)UDS                                                                | lorazepam (2 mg, IV), IVF                           | Müller et al. (2010a,b)        |
| Russia  | 3  | 2M 1F  | 22 ± 1 | Tropical Synergy                                       | ~1 g each          | Reddened conjunctivae, tachycardia, anxiety, paranoia, hallucinations, short-term memory & sense of time impairment                                                                                                                                                               | UNK                  | UNK      | (–)UDS, (+)JWH-018 metabolites (urine)                                | UNK                                                 | Sobolevsky et al. (2010)       |
| USA     | 1  | F      | 17     | JWH-018                                                | UNK (sm)           | “violent” & “crazy”, hallucinations, lower extremities numbness, muscle twitches, elevated pulse, dilated pupils                                                                                                                                                                  | 15 min               | 2 h      | (+)THC (urine), SC not tested                                         | lorazepam (2 mg, IV)                                | Vearrier and Osterhoudt (2010) |
| USA     | 1  | M      | 20     | Spice                                                  | UNK (sm)           | Anxiety, tachycardia, diaphoresis                                                                                                                                                                                                                                                 | UNK                  | UNK      | (–)UDS, SC not tested                                                 | supportive care                                     | Benford and Caplan (2011)      |
| USA     | 11 | 10M 1F | 15–19  | UNK                                                    | UNK (sm)           | Euphoria (11), irritability (4), anxiety (3), numbness (2), anger (1), sadness (1), memory impairment (11), change of auditory (1) & visual (5) perception, paranoia (2), palpitations (3), muscle trembles (1) & weakness (1), blackouts (1), restlessness (1), stimulation (10) | UNK                  | UNK      | UNK                                                                   | UNK                                                 | Castellanos et al. (2011)      |
| Germany | 13 | 12M 1F | 14–28  | Spice, Smoke, Jamaican Gold, Monkees-go bananas, Ninja | UNK (sm)           | Shaking, acute psychosis, seizures, muscle jerking, muscle pains, hypokalemia                                                                                                                                                                                                     | UNK                  | UNK      | (+)JWH-018(7), JWH-081(4), JWH-122(4); all serum, concentration UNK   | UNK                                                 | Hermanns-Clausen et al. (2011) |
| USA     | 10 | M      | 21–25  | UNK                                                    | UNK (sm)           | Auditory hallucinations (4), paranoid delusions (9), odd or flat affect (6), blocked thoughts (4), disorganized speech (6) & behavior (7), alogia (3), suicidal ideation (4), insomnia (6), psychomotor retardation (6) & agitation (3), anxiety (2)                              | UNK                  | UNK      | (+)THC(4), (–)UDS(6), SC not tested                                   | antipsychotics (7), hospitalization (6–10d)         | Hurst et al. (2011)            |
| USA     | 1  | M      | 23     | Spice                                                  | UNK (sm)           | Nonsensical speech, paranoia, disorganized thoughts                                                                                                                                                                                                                               | <48 h                | <72 h    | (–)UDS                                                                | psychiatric referral, supportive care               | Johnson et al. (2011)          |
| USA     | 1  | M      | 48     | JWH-018 (powder)                                       | UNK (po with ETOH) | Seizure, tachycardia, refractory supraventricular tachycardia (1 day later)                                                                                                                                                                                                       | 30 min               | <48 h    | (–)UDS, 74.3 µg/L JWH-018 pentanoic (urine)                           | lorazepam (IV), electrocardioversion; ET intubation | Lapoint et al. (2011)          |
| Italy   | 10 | UNK    | 14–55  | Spice, N-joy, Forest Green (contained JWH-122)         | UNK                | Agitation (7), confusion (6), hallucination (4), dyspnea (1), coma (2), seizure (1), mydriasis (2), xerostomia (2), vertical nystagmus (1), psychomotor agitation (2), vomiting (1)                                                                                               | UNK                  | <24 h    | (+)JWH-018 (blood), (+)JWH-250 (blood & urine); number of samples UNK | UNK                                                 | Locatelli et al. (2011)        |
| USA     | 1  | M      | 16     | K2                                                     | UNK (sm)           | Chest pain for 3d, diagnosed with elevated ST-segment & troponin (25 µg/L)                                                                                                                                                                                                        | 1d after sm          | 3d       | (+)THC, SC not tested                                                 | supportive care, coronary angiography               | Mir et al. (2011)              |
|         | 1  | M      | 16     | K2                                                     | UNK (sm)           | Intermittent chest pain × 3d lasting ~30 min, diagnosed with elevated ST-segment & troponin (11.6 µg/L)                                                                                                                                                                           | within 1 wk after sm | UNK      | (–)UDS, SC not tested                                                 | supportive care, coronary angiography               |                                |
|         | 1  | M      | 16     | K2                                                     | UNK (sm)           | Intermittent chest pain × 3d lasting 1–2 h, elevated troponin (12 µg/L)                                                                                                                                                                                                           | 4d after sm          | 1 wk     | (+)THC, (+)JWH-018, JWH-073 metabolites (urine)                       | supportive care                                     |                                |

|     |   |     |        |                                                                  |               |                                                                                                                               |                                  |        |                                                                       |                                                              |                                         |
|-----|---|-----|--------|------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------|-----------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------|
| USA | 9 | UNK | UNK    | UNK                                                              | UNK           | Tachycardia, hypokalemia, agitation/irritability, hallucination, pallor, nausea, mydriasis                                    | UNK                              | 8–24 h | (–)UDS, (+)JWH-018, JWH-073 metabolites (urine)                       | BDP, IVF, anti-emetics, potassium supplement supportive care | <a href="#">McCain et al. (2011)</a>    |
| USA | 3 | UNK | UNK    | Bad Mojo                                                         | UNK (sm)      | psychosis                                                                                                                     | UNK                              | >12 h  | UNK                                                                   |                                                              | <a href="#">Rodgman et al. (2011)</a>   |
| USA | 2 | F   | 20, 22 | Banana Cream Nuke                                                | ½ packet (sm) | Anxiety, palpitations/tachycardia                                                                                             | “shortly after smoking”          | 1 h    | (–)UDS, SC not tested                                                 | supportive care                                              | <a href="#">Schneir et al. (2011)</a>   |
| USA | 1 | M   | 25     | UNK                                                              | UNK (sm)      | “Eyes-crossed & flailing arms”, unresponsive to verbal stimuli, dilated pupils                                                | 45 min                           | 3 h    | (–)UDS, (+)JWH-018 metabolites (urine)                                | lorazepam (4 mg, IV), IVF                                    | <a href="#">Simmons et al. (2011)</a>   |
|     | 1 | M   | 23     | UNK                                                              | UNK (sm)      | Unresponsive, muscle spasm, depressed breathing                                                                               | UNK                              | <24 h  | (–)UDS, (+)JWH-018, JWH-073 metabolites (urine)                       | airway management, ICU admission, 5 mg haloperidol           |                                         |
|     | 1 | M   | 19     | UNK                                                              | UNK (sm)      | Paranoia, delusions, short-term memory impairment                                                                             | 1 h                              | <24 h  | (–)UDS, (+)JWH-018 & JWH-073 metabolites                              | supportive care                                              |                                         |
| USA | 1 | M   | 17     | K9 Pure Fire (contained JWH-073, JWH-018)                        | UNK (sm)      | Hallucinations, dizziness, difficulty breathing, tachycardia, chest pressure that lasted for 3 days                           | 10 min                           | <24 h  | (–)UDS, SC not tested                                                 | nitroglycerin, supportive care                               | <a href="#">Young et al. (2011)</a>     |
| USA | 1 | M   | 19     | Space                                                            | UNK (sm)      | Paranoia, hallucinations, agitation, tachycardia, hyperglycemia                                                               | 2 h                              | <6 h   | (–)UDS, SC not tested                                                 | lorazepam (2 mg, IV)                                         | <a href="#">Bebarta et al. (2012)</a>   |
|     | 1 | F   | 19     | Space                                                            | UNK (sm)      | Mild drowsiness, short-term memory impairment, hyperglycemia                                                                  | UNK                              | <6 h   | (–)UDS, (+)APAP, (+)DXM, (+)doxylamine, (+)levorphenol, SC not tested | supportive care                                              |                                         |
| USA | 1 | M   | 23     | Spice                                                            | UNK (sm)      | Anxiety, agitation, breathing difficulty, hyperventilation, tachycardia, injected sclera                                      | UNK                              |        | (–)UDS, SC not tested                                                 | lorazepam, IVF, anti-emetic                                  | <a href="#">Cohen et al. (2012)</a>     |
|     | 1 | F   | 16     | K2                                                               | UNK (sm)      | Catatonia, tachycardia, vertical nystagmus                                                                                    | UNK                              | <24 h  | (–)UDS, SC not tested                                                 | diphenhydramine (50 mg, IV), lorazepam (2 mg IV × 2 dose)    |                                         |
|     | 1 | M   | 18     | Spice                                                            | UNK (sm)      | Headache, dizziness, profuse sweating, agitation, aggression, restlessness, tachycardia, hyperventilation                     | UNK                              | <24 h  | (–)UDS, SC not tested                                                 | diphenhydramine (50 mg, IV), lorazepam (2 mg, IV)            |                                         |
|     | 1 | M   | 16     | Spice                                                            | UNK (sm)      | Disorientation, agitation, slowed speech                                                                                      | UNK                              | <24 h  | (–)UDS, SC not tested                                                 | lorazepam (4 mg IV), IVF                                     |                                         |
| USA | 1 | M   | 17     | K2                                                               | UNK (sm)      | Dizziness, confusion, lethargy, vomiting, tachycardia                                                                         | “immediately after 1 inhalation” | UNK    | (–)UDS, SC not tested                                                 | IVF, naloxone (2 mg, IV), supportive care                    | <a href="#">Faircloth et al. (2012)</a> |
| USA | 3 | M   | 20–30  | K2, XXX, K2 Blond, Black Box, Smoke n’ Skulls, Zombie, Blueberry | 3 g/d (sm)    | Similar (1) or elevated “high” than cannabis (2)                                                                              | UNK                              | UNK    | UNK                                                                   | UNK                                                          | <a href="#">Gunderson et al. (2012)</a> |
| USA | 1 | F   | 19     | Bayou Blaster®                                                   | UNK (sm)      | Jerking motions of extremities, agitation, altered mental status, somnolence, tachycardia                                     | “immediately after smoking”      | <3 h   | (–)UDS, SC not tested                                                 | admitted to mental health ward; discharged after 4 days      | <a href="#">Harris and Brown (2012)</a> |
|     | 1 | M   | 17     | Humboldt Gold                                                    |               | Agitation, hallucinations, myoclonic jerking, aggression, tachycardia, flushed skin, dilated pupils, “inappropriate laughter” | UNK                              | <3 h   | (–)UDS, SC not tested                                                 |                                                              |                                         |

**Table 2** (Continued)

| Country | N  | Sex    | Age   | SC Brand                                                                                                                | Dose (Route) | Symptoms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Time onset                  | Duration    | Laboratory results (matrix)                                                                                                                                                                             | Treatment                                                                                                                          | Reference                       |
|---------|----|--------|-------|-------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| USA     | 1  | M      | 17    | K2                                                                                                                      | UNK (sm)     | Hypertonia, apnea, cyanosis, confused, swollen red eyes, tachycardia, chest & back pain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | UNK                         | <12 h       | (–)UDS, SC not tested                                                                                                                                                                                   | adenosine (6 mg, IV), APAP                                                                                                         | Heath et al. (2012)             |
|         | 1  | M      | 15    | K2                                                                                                                      | UNK (sm)     | Loss of consciousness, tachycardia, headache, fatigue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | UNK                         |             | (–)UDS, SC not tested                                                                                                                                                                                   | IVF                                                                                                                                |                                 |
| Germany | 29 | 25M 4F | 14–30 | Bonzai, Jamaican Gold, Lava Red, Maya, Monkees go bananas tropical car perfume, Ninja Strong, OMG, Spice, Smoke, Space, | UNK (sm)     | Poisoning severity score (PSS) 1 ( $n=9$ ): drowsiness, vertigo, ataxia, restlessness, paraesthesia, mild visual or auditory hallucinations mild muscular tenderness or pain, tachycardia, mild change in blood pressure, vomiting, diarrhea, abdominal pain, mild hypoglycemia, mild electrolyte imbalance, short-term hypothermia<br>PSS2 ( $n=18$ ): unconsciousness, brief apnea or slowed breathing, confusion, agitation, hallucination, delirium, seizures, visual and auditory hallucinations, dystonia, rhabdomyolysis or chest pain, sinus brady or tachycardia, irregular EKG, prolonged coughing, bronchospasm, difficulty breathing, vomiting, diarrhea, abdominal pain, electrolyte imbalance, moderate hypoglycemia, and prolong hypothermia<br>Seizure, unresponsive, aggression, “abnormal” speech, slow & shallow breathing, loss of consciousness | 1–20 h                      | UNK         | 2.3 µg/L CP47,497-C8 (1), <0.1–13 µg/L JWH-018 (8), 0.11 µg/L JWH-073(1), 1.2–42 µg/L JWH-081(7), 0.17–40 µg/L(11), 2.5–190 µg/L JWH-210(11), 0.1–1.1 JWH-250(4), 0.2 µg/L AM694; all measured in serum | Supportive care (29), BDP (8), IVF (5), anti-emetics (2), potassium supplement (5), neuroleptics (1), psychiatric care (1), ET (1) | Hermanns-Clausen et al. (2013b) |
| USA     | 1  | M      | 19    | Climax                                                                                                                  | “pinch” (sm) | Seizure, unresponsive, aggression, “abnormal” speech, slow & shallow breathing, loss of consciousness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | UNK                         | UNK         | (–)UDS, SC not tested                                                                                                                                                                                   | supportive care                                                                                                                    | Jinwala and Gupta (2012)        |
|         | 1  | M      | 15    | Silver-K2                                                                                                               | UNK (sm)     | Loss of consciousness, shallow breathing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | UNK                         | UNK         | (–)UDS< SC not tested                                                                                                                                                                                   | ET intubation, supportive care                                                                                                     |                                 |
| USA     | 1  | F      | 18    | KS                                                                                                                      | UNK (sm)     | Panic attack, paranoia, chest pain, hyperventilation, nausea                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | UNK                         | UNK         | (–)UDS, SC not tested                                                                                                                                                                                   | supportive care                                                                                                                    | McGuiness and Newell (2012)     |
| USA     | 1  | M      | 20    | Black Mamba                                                                                                             | UNK (sm)     | Tonic–clonic seizures, dry skin, drowsiness, elevated pulse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | “immediately after smoking  | <3 h        | (+)AM2201 metabolites                                                                                                                                                                                   | IVF, supportive care                                                                                                               | McQuade et al. (2012)           |
| USA     | 1  | M      | 59    | Spice                                                                                                                   | 1.5 g/d (sm) | “Flashbacks” combat-related trauma, re-admitted 3wk later for hallucinations, then 2d later, total 3 admissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | UNK                         | <24 h/visit | (–)UDS, SC not tested                                                                                                                                                                                   | BDP, gabapentin (400 mg QID), hydroxyzine (25 mg TID PRN), apiprazole (10 mg qD), benztropine (1 mg BID), bupropion (150 mg BID)   | Peglow et al. (2012)            |
| USA     | 1  | M      | 48    | Spice                                                                                                                   | 3 g (sm)     | Tonic–clonic seizures, tachycardia, diaphoresis, mydriasis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | UNK                         | <24 h       | (–)UDS, 140 mg/dL BAC (serum), (+)JWH-018 metabolites (urine)                                                                                                                                           | lorazepam (4 mg IV)                                                                                                                | Pant et al. (2012)              |
| USA     | 1  | M      | 19    | Happy Tiger (contained JWH-018, JWH-081, JWH-250, AM2201)                                                               | UNK (sm)     | Convulsions, vomiting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | “immediately after smoking” | <24 h       | (–)UDS, SC not tested                                                                                                                                                                                   | midazolam (5 mg IV), supportive care                                                                                               | Schneir and Baumbacher (2012)   |

|           |    |        |       |                                                                                      |                      |                                                                                                                                                                                  |                    |       |                                                                                                                                                                                                                                        |                                                         |                                                                    |
|-----------|----|--------|-------|--------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------|
| USA       | 1  | M      | 20    | K2                                                                                   | UNK (sm)             | Agitation, confusion, suicidal ideation, self-inflicted wounds, hyperventilation, tachycardia                                                                                    | UNK                | <24 h | (–)UDS, SC not tested                                                                                                                                                                                                                  | supportive care                                         | <a href="#">Thomas et al. (2012)</a>                               |
| USA       | 1  | M      | 48    | K2 Summit                                                                            | 0.3 g (po with milk) | Sedated, nauseous, “detached,” flushed, loss of consciousness, tonic–clonic seizure, tachycardia, depressed breathing, hyperthermia, supraventricular tachycardia                | 45 min             | <48 h | (–)UDS, SC not tested                                                                                                                                                                                                                  | lorazepam (IV), electrocardioversion, ET intubation     | <a href="#">Tofighi and Lee (2012)</a>                             |
| Hong Kong | 1  | M      | 36    | K2                                                                                   | 0.5 g/d (sm)         | Agitation, profuse sweating, tachycardia, delusion, elevated blood pressure                                                                                                      | UNK                | <24 h | (–)UDS, (+)DXM, (+)ephedrine, (+)promethazine, SC not tested                                                                                                                                                                           | midazolam (IM)                                          | <a href="#">Tung et al. (2012)</a>                                 |
| USA       | 1  | M      | 21    | MadHatter, Kite, Scooby Snax                                                         | UNK (sm)             | Fainted while driving, tachycardia, elevated blood pressure, dyspnea & hyperventilation                                                                                          | UNK                | UNK   | (+)THC (urine), 0.75 µg/L AM2201, JWH-122, JWH-210 (blood), (+)AM2201 & JWH-018 metabolites (urine)                                                                                                                                    | ICU admission, airway management, antibiotics, steroids | <a href="#">Alhadi et al. (2013)</a>                               |
| USA       | 1  | F      | 22    | K2                                                                                   | UNK (sm)             | Palpitations, dyspnea, “angor animi,” dysarthria, difficulty standing, drowsiness, inattention, left face & hemi-body weakness, & hemianesthesia, diagnosed with ischemic stroke | “while smoking”    | UNK   | (+)THC, BDP, & salicylates (urine)                                                                                                                                                                                                     | supportive care                                         | <a href="#">Bernson-Leung et al. (2013)</a>                        |
|           | 1  | F      | 26    | Peak Extreme                                                                         | UNK (sm)             | Confirmed by CT scan left facial weakness, left-sided numbness, dysfluency, hemi-anesthesia, left visual neglect, diagnosed with ischemic stroke confirmed by CT scan            | <24 h              | UNK   | (–)UDS                                                                                                                                                                                                                                 | warfarin, supportive care                               |                                                                    |
| USA       | 1  | M      | 20    | Spice                                                                                | UNK (sm)             | Uncommunicative, unable to follow instruction, combatant                                                                                                                         | UNK                | UNK   | (–)UDS, SC not tested                                                                                                                                                                                                                  | IVF, lorazepam (2 mg) admission, supportive care        | <a href="#">Berry-Caban et al. (2013)</a>                          |
| USA       | 1  | M      | 20    | Spice                                                                                | UNK (sm)             | Nausea, vomiting for 2 d; history of smoking SC within the last few weeks                                                                                                        | 48 h               | UNK   | (–)UDS, SC not tested                                                                                                                                                                                                                  | renal biopsy, inpatient admission, supportive care      | <a href="#">Bhanushali et al. (2013)</a>                           |
|           | 1  | M      | 23    | Spice                                                                                | UNK (sm)             | Nausea, vomiting for 2 d; history of SC within few weeks                                                                                                                         | 48 h               | UNK   | (–)UDS, SC not tested                                                                                                                                                                                                                  | renal biopsy, inpatient admission, supportive care      |                                                                    |
|           | 1  | M      | 26    | Spice                                                                                | UNK (sm)             | Nausea, vomiting, diarrhea, lower abdominal pain × 2 d; history of SC intake × 2 yr, but changed “supplier” in last week                                                         | 48 h               | UNK   | (–)UDS, SC not tested                                                                                                                                                                                                                  | renal biopsy, inpatient admission, supportive care      |                                                                    |
|           | 1  | M      | 30    | Spice                                                                                | UNK (sm)             | Nausea, vomiting, diarrhea, abdominal pain × 3 d; history of SC intake × 1 yr, but changed “supplier” in few wks.                                                                | 72 h               | UNK   | (–)UDS, SC not tested                                                                                                                                                                                                                  | inpatient admission, supportive care                    |                                                                    |
| USA       | 16 | 15M 1F | 15–33 | Blueberry, Clown Loyal, Flame 2.0, Mad Monkey, Mr. Happy, Phantom Wicked, Spice Gold | UNK (sm)             | Nausea (15), vomiting (15), abdominal (8) & flank (4) pain, back pain (2)                                                                                                        | “within h or days” | UNK   | 42 µg/L XLR-11 pentanoic acid (blood, n = 1), 35–35 µg/L XLR-11, 38–102 µg/L XLR-11 pentanoic acid metabolite (serum, n = 2), 6 µg/L UR-144 (serum, n = 1), 400–529 µg/L XLR-11 pentanoic acid (urine, n = 2); SC not tested for n = 9 | renal biopsy(8), hemodialysis (5), corticosteroid (4)   | <a href="#">Centers for Disease Control and Prevention (2013a)</a> |

Table 2 (Continued)

| Country     | N  | Sex     | Age     | SC Brand                                                                                                                                                | Dose (Route) | Symptoms                                                                                                                                                                                                                                            | Time onset              | Duration | Laboratory results (matrix)                                                                                                                   | Treatment                                                        | Reference                                                          |
|-------------|----|---------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|
| USA         | 17 | 11M, 6F | 13–60   | Black Mamba, Crazy Monkey, Crazy Clown, Dead Man Walking, Funky Monkey, Sexy Monkey, SinX, Spice, TenX, Twilight, 3X Crazy Clown (contained ADB-PINACA) | UNK (sm)     | Elevated blood pressure (81), tachycardia (73), somnolence(45), aggressive or violent behavior (40), agitation (40), confusion (32)                                                                                                                 | UNK                     | UNK      | UNK                                                                                                                                           | inpatient (7) & ICU(10) admission                                | <a href="#">Centers for Disease Control and Prevention (2013b)</a> |
| USA         | 22 | 18M 4F  | 16–57   |                                                                                                                                                         | UNK          | Hyperglycemia (13), hypokalemia (9), acidosis (7), tachycardia (13), nausea/vomiting (8), confusion/disorientation (7), aggression (7), somnolence/unresponsiveness (7), seizures (3), pneumonia(2), rhabdomyolysis (1), myocardial infarction (1). | UNK                     | UNK      | (+)ADB-PINACA (n = 5, serum)                                                                                                                  | ICU admission (6), assistant ventilation (5)                     | <a href="#">Centers for Disease Control and Prevention (2013c)</a> |
| Switzerland | 1  | M       | 31      | Samurai King                                                                                                                                            | 300 mg (sm)  | Agitation, aggression, anxiety, confusion, panic, vomiting, dilated pupils, tachycardia, elevated blood pressure, hyperglycemic                                                                                                                     | “few min”               | <3 h     | (–)UDS, 49 µg/L MAM2201 (plasma) after 1 h                                                                                                    | supportive care                                                  | <a href="#">Derungs et al. (2013)</a>                              |
| USA         | 1  | M       | 26      | Spice (contained JWH-018)                                                                                                                               | UNK (sm)     | Dysarthria, expressive aphasia, right face & arm weakness, incoherent speech, tachycardia                                                                                                                                                           | 50 min                  |          | (+)THC (urine), SC not tested                                                                                                                 | brain MRI revealed cerebral infarct & patient treated for stroke | <a href="#">Freeman et al. (2013)</a>                              |
|             | 1  | F       | 19      | Spice (contained JWH-018)                                                                                                                               |              | Loss of consciousness, vomiting, altered mental status, & jerking movements of extremities, aphasia, sensory loss, right hemiplegia, tachycardia                                                                                                    | “a few min” after sm    |          | (+)THC (urine)                                                                                                                                | brain MRI revealed clot & cerebral infarct                       |                                                                    |
| New Zealand | 17 | 10M 7F  | 26 mean | K2                                                                                                                                                      | UNK          | Paranoia, thought disorder, disorganized behavior, anxiety, depression, suicidal ideation (3), homicidal ideation (1)                                                                                                                               | UNK                     | UNK      | UNK                                                                                                                                           | inpatient psychiatric unit admission                             | <a href="#">Glue et al. (2013)</a>                                 |
| Germany     | 1  | M       | 19      | Bonzai                                                                                                                                                  | UNK (sm)     | Tonic–clonic seizures, unresponsive, vomiting                                                                                                                                                                                                       | <24 h                   | <72 h    | 0.4 µg/L JWH-018, 230 µg/L JWH-122, 7.8 µg/L JWH-210 (serum), 39 µg/L THC (urine), (+) JWH-018, JWH-073, JWH-122, JWH-210 metabolites (urine) | midazolam, ventilation, supportive care                          | <a href="#">Hermanns-Clausen et al. (2013a)</a>                    |
|             | 1  | M       | 17      | NS                                                                                                                                                      | UNK (sm)     | Somnolence, tachycardia, mydriasis, unequal pupil size, retrograde amnesia                                                                                                                                                                          | “shortly after smoking” | <12 h    | 0.2 µg/L MAM2201, 0.2 µg/L UR-144 (serum), (+)MAM2201, UR-144, JWH-018 metabolites (urine)                                                    | supportive care                                                  |                                                                    |
|             | 1  | M       | 17      | Jamaican                                                                                                                                                | UNK (sm)     | Nausea, mild agitation, trembling, “laugh attacks,” confusion, somnolence, mydriasis, tachycardia, hyperglycemia                                                                                                                                    | “few min”               | <6 h     | 42 µg/L JWH-081 (serum), (+)JWH-018, JWH-073, JWH-081 metabolites (urine)                                                                     | IVF, K+ supplement, supportive care                              |                                                                    |
|             | 1  | M       | 20      | Lava                                                                                                                                                    | UNK (sm)     | Skin pallor, vomiting, “rolled his eyes,” tachycardia, somnolence                                                                                                                                                                                   |                         |          | 15.0 µg/L JWH-122 & metabolite, JWH-018 metabolites (serum)                                                                                   | BDP, IVF, potassium supplement                                   |                                                                    |



|     |   |    |    |                                                                    |            |                                                                                                                                        |                    |       |                                                                                                               |                                                                                    |                                              |
|-----|---|----|----|--------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------|
| USA | 1 | M  | 30 | Scooby Snacks (contained JWH-018, JWH-073, JWH-122, AM2201, AM694) | UNK (sm)   | Abdominal pain, nausea, vomiting (intermittent); history of ED visit 3d prior for the same symptoms                                    | UNK                | UNK   | (–)UDS, (+)JWH-018, JWH-073, AM2201 metabolites (urine)                                                       | IVF, ondansetron (IVF), discharged with promethazine                               | <a href="#">Hopkins and Gilchrist (2013)</a> |
| USA | 1 | F  | 28 | K2 (contained JWH-018 & JWH-122)                                   | UNK (sm)   | Nausea, left-side hemiplegia & paralysis, incoherent & slurred speech, confusion                                                       | UNK                | UNK   | UNK                                                                                                           | brain MRI revealed multiple embolism                                               | <a href="#">Korya et al. (2013)</a>          |
| USA | 1 | 18 | M  | Jamaican Extreme                                                   | UNK (sm)   | Unable to follow instruction, slowed movement, confused & disoriented, slurred & babbling speech, slowed breathing, dilated pupils     | UNK                | UNK   | 4.6 µg/L AM2201, 0.2 µg/L JWH-018 (blood)                                                                     | UNK, DUID case                                                                     | <a href="#">Musschoff et al. (2013)</a>      |
|     | 2 | 14 | F  | BooM                                                               | UNK (sm)   | Confused, slurred & babbling speech, disoriented, “blunt” mood                                                                         | UNK                | UNK   | 0.4 & 0.8 µg/mL JWH-210, 0.3 µg/mL JWH-122 (blood)                                                            | UNK, DUID case                                                                     |                                              |
|     | 1 | 20 | M  | NS                                                                 | UNK        | “Vestibular disorder”, impaired fine motor skills, dilated pupils, delayed reaction, “blunt” mood                                      | UNK                | UNK   | 0.3 µg/L AM2201, 1.7 µg/L JWH-019, 7.6 µg/L JWH-122, 4.4 µg/L JWH-210 (blood)                                 | UNK, DUID case                                                                     |                                              |
|     | 1 | 29 | M  | BooM, OMG                                                          | UNK        | Dilated pupils with delayed reaction to light, “dizzy mind & retarded behavior”                                                        | UNK                | UNK   | 1.0 µg/L JWH-122, 6.2 µg/L JWH-210 (blood)                                                                    | UNK, DUID case                                                                     |                                              |
|     | 1 | 21 | M  | UNK                                                                | UNK        | Delayed reaction & movement, anxiety, constricted pupils with no reaction to light, “dizzy mind,” depressed mood                       | UNK                | UNK   | 0.5 µg/L JWH-018, 0.3 µg/L JWH-122, 0.7 µg/L JWH-210 (blood)                                                  | UNK, DUID case                                                                     |                                              |
|     | 1 | 21 | M  | UNK                                                                | UNK        | No signs of impairment except for smell of alcohol in breath                                                                           | UNK                | UNK   | 1.74% BAC, 1.1 µg/L JWH-307 (blood)                                                                           | UNK, DUID case                                                                     |                                              |
|     | 1 | 22 | M  | UNK                                                                | UNK        | Delayed movement, anxiety, lack of apathy, “inert”, delayed pupils’ reaction to light                                                  | UNK                | UNK   | 1.9 µg/L JWH-018, 28 µg/L JWH-122, 2.5 µg/L JWH-210, <0.1 µg/L for AM2201, JWH-307, MAM2201, & UR-144 (blood) | UNK, DUID case                                                                     |                                              |
| USA | 1 | F  | 22 | Mr. Nice Guy                                                       | 1 g/d (sm) | Cramping in extremities, sweats & chills, cravings, headache, insomnia, vivid dreams, anorexia, anxiety, tachycardia, hyperventilation | 6 d prior ED visit | <1 wk | UNK                                                                                                           | lorazepam (2 mg IV), IVF, BDP                                                      | <a href="#">Nacca et al. (2013)</a>          |
|     | 1 | M  | 20 | Mr. Nice Guy                                                       | 3 g/d (sm) | Chest pain, palpitations, dyspnea, headache, diaphoresis, tremors, tachycardia                                                         | 6 d prior ED visit |       | UNK                                                                                                           | inpatient admission, BDP, hydroxyzine, diphenhydramine; discharged with quetiapine |                                              |

Table 2 (Continued)

| Country | N  | Sex    | Age   | SC Brand                                             | Dose (Route) | Symptoms                                                                                                                                            | Time onset | Duration | Laboratory results (matrix)                                                                                                                                                                                                                                                                                                                                                   | Treatment                  | Reference               |
|---------|----|--------|-------|------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------|
| USA     | 1  | M      | 26    | Mr. Happy (contained 69 mg/g XLR-11, 61 mg/g UR-144) | UNK (sm)     | Abdominal pain, nausea, vomiting, lower back pain.                                                                                                  | <24 h      |          | (–)UDS, 529 µg/L XLR-11 pentanoic acid (urine), 36 µg/L XLR-11, 6 µg/L UR-144, 102 µg/L XLR-11 pentanoic acid (serum) initial; 34 µg/L XLR-11, 3 µg/L UR-144, 62 µg/L XLR-11 pentanoic acid after 3.5 h after initial test                                                                                                                                                    | inpatient admission for 6d | Thornton et al. (2013)  |
| Norway  | 16 | 15M 1F | 17–41 | UNK                                                  | UNK          | Mild to moderate impairment                                                                                                                         | UNK        | UNK      | <0.1 µg/L THC (n = 10), 0.1–1.3 µg/L AM2201 (n = 3), 0.1–0.5 µg/L JWH-018 (n = 3), 0.5–1.7 µg/L JWH-122 (n = 2), 0.5 µg/L JWH-250 (n = 1), 1.0 µg/L RCS-4 (n = 1); <0.1–1.0 µg/L AMP (n = 4), <0.1–1.9 µg/L MAMP (n = 7), <0.1–2.4 µg/L BDP (n = 12), 0.11 ppm ETOH (n = 1), <0.1 µg/L LSD (n = 2), <0.1 µg/L ketamine (n = 1); all analytes confirmed in whole blood samples | UNK, DUID cases            | Tuv et al. (2014)       |
| USA     | 12 | M      | 18–31 |                                                      |              | Vertical nystagmus (1), tachycardia (9), elevated blood pressure (2), body & extremities tremors (6), eye tremors (5), swaying or unsteady gait (5) | UNK        | UNK      | 0.4–4.0 µg/L (n = 7), 0.1–1.1 µg/L JWH-018 (n = 6), <0.1–0.1 µg/L JWH-081, <0.1–2.5 µg/L JWH-122 (n = 3), <0.1–0.1 µg/L JWH-210 (n = 4), 0.4–2.7 µg/L JWH-250 (n = 2)                                                                                                                                                                                                         | UNK, DUID cases            | Yeakel and Logan (2013) |

Abbreviations: ADB-PINACA [N-(1-Amino-3,3-dimethyl-1-oxobutan-2-yl)-1-pentyl-1H-indazole-3-carboxamide], AM2201 [N-(5-Fluoropentyl)-3-(1-naphthoyl)indole], AMP (amphetamine), APAP (acetaminophen), BDP (benzodiazepines), BID (twice a day), DXR (dextromethorphan), DUID (driving under the influence of drugs), ECG (electrocardiogram), ED (emergency department), ET (endotracheal intubation), ETOH (ethanol), ICU (intensive care unit), IVF (intravenous fluids), IV (intravenous), JWH-018 [Naphthalen-1-yl-(1-pentylindol-3-yl)methanone], JWH-019 [(1-Hexylindol-3-yl)naphthalene-1-ylmethanone], JWH-073 [naphthalen-1-yl-(1-butyndol-3-yl)methanone], JWH-081 [4-Methoxy-1-naphthalen-1-yl-(1-pentylindol-3-yl)methanone], JWH-122 [1-Pentyl-3-(1-(4-methylnaphthoyl))indole], JWH-210 [4-Ethylindol-3-yl-(1-pentylindol-3-yl)methanone], JWH-250 [1-Pentyl-3-(2-methoxyphenylacetyl)indole], JWH-307 [(5-(2-Fluorophenyl)-1-pentylpyrrol-3-yl)-1-(naphthalenyl)methanone], MAM2201 [1-(5-Fluoropentyl)-3-(4-methyl-1-naphthoyl)indole], MRI (magnetic resonance imaging), qD (every day), QID (four times a day), po (oral), RCS-4 [4-Methoxyphenyl-(1-allyl-1H-indol-3-yl)methanone], sm (smoked), THC (delta-9-tetrahydrocannabinol), TID (three times a day), UDS (urine drug screen), UNK (unknown), UR-144 [(1-Pentylindol-3-yl)-(2,2,3,3-tetramethylcyclopropyl)methanone], XLR-11 [(1-(5-Fluoropentyl)-1H-indol-3-yl)(2,2,3,3-tetramethylcyclopropyl)methanone].

apparently healthy men (20–30 years) with no prior psychiatric history developed new onset psychosis after smoking SC, including paranoia, thought disorder, and suicidal ideation (5 patients) (Hurst et al., 2011; Thomas et al., 2012; Van Der Veer and Friday, 2011). Treatment required psychiatric hospitalization and anti-psychotic medications. The episode lasted up to one week in 8 patients, two weeks in 3 patients, and more than 5 months in 3 patients.

Fifteen men hospitalized in a New Zealand psychiatric ward were interviewed about SC use (Every-Palmer, 2011). All had a history of cannabis intake, while the majority (13) reported smoking SC (JWH-018) within the past year and experiencing rapid onset of psychoactive effects after smoking SC alone. Of the 13 SC smokers, 9 admitted to experiencing psychotic symptoms. Although none reported experiencing withdrawal, 5 patients experienced psychotic relapse after smoking SC for a day and effects lasted as long as “several weeks.” A retrospective audit of SC-related admissions in a New Zealand acute psychiatric ward reported 17 patients with severe psychotic symptoms hospitalized for up to 13.1 days (Glue et al., 2013).

One 59-year old man with a history of post-traumatic stress disorder and polydrug abuse experienced “flashbacks” immediately after smoking SC (Peglow et al., 2012). He had three hospital admissions over 60 days, recovering within 24 h with each subsequent visit. All his drug toxicological screens were negative for alcohol, THC, cocaine, opiates, barbiturates, benzodiazepines, and PCP. His episodes ceased after discontinuing SC intake. Another case involved a 25-year old man with a history of recurrent psychosis who developed anxiety and paranoid psychosis with auditory command hallucinations after smoking 3 g “Spice” on three occasions, despite being on stable anti-psychotic treatment (amisulpiride 800 mg daily) for 2 years (Müller et al., 2010a,b). Symptoms persisted for one month and required psychiatric hospitalization. The patient described previous similar psychotic exacerbations induced by smoking cannabis.

**3.3.5. Withdrawal.** We identified 3 peer-reviewed articles addressing SC withdrawal. A 20-year-old man who smoked “Spice Gold” 3 g/day for 8 months was hospitalized about 1.5 days after last use with a severe withdrawal syndrome, including increased craving, restlessness, nightmares, tachycardia (maximum heart rate 125 bpm), hypertension (180/90), nausea, sweating, and muscle twitches. The syndrome resolved within one week with symptomatic treatment (Zimmermann et al., 2009). A 22-year-old woman smoking 3 g/day SC presented to the ED complaining of severe anxiety, “vivid” dreams, headache, cramping of extremities, “sweats and chills”, anorexia, and craving 6 days after last use (Nacca et al., 2013). She was discharged within 3 h of receiving i.v. saline and 2 mg lorazepam. A 20-year-old man, with a history of smoking “Mr. Nice Guy” for 18 months, ceased smoking 6 days prior to ED presentation for headache, chest pain, profuse sweating, and body tremors. Prior to his ED visit, he attempted to alleviate symptoms by smoking cannabis, which was ineffective, but taking his roommate’s quetiapine provided relief. Benzodiazepines did not alleviate his symptoms, and after inpatient admission, subsequent hydroxyzine and diphenhydramine administrations also were unsuccessful. The patient’s symptoms subsided after the physician administered 50 mg quetiapine, and he was released with an unspecified quetiapine dose.

A 23-year-old man, admitted to a German inpatient detoxification clinic for SC withdrawal symptoms, underwent positron emission tomography (PET) scans with the dopamine ( $D_2/D_3$ ) receptor ligand [ $^{18}F$ ] fallypride one day after admission and after one week of SC abstinence (Rominger et al., 2013). Compared to controls (three healthy males ages 20, 21, and 25), the patient’s first PET scan showed globally decreased  $D_2/D_3$  binding, with the lateral temporal cortex and hippocampus least (–15%) and most (–62%)

affected, respectively. The second scan showed marked increases in several areas, with the majority returning to control values. Effects from cannabis were considered unlikely, as the patient’s urine drug screen was negative for cannabinoids.

It is difficult to directly compare SC and cannabis withdrawal symptoms because data are limited and patients often had a history of smoking cannabis before and/or after SC. However, many of the symptoms reported by patients after cessation of SC use are also reported during cannabis withdrawal, e.g., disturbed sleep and dreaming, anxiety, craving, nausea, muscle twitching or cramping, and chills (Levin et al., 2010; American Psychiatric Association, 2013).

**3.3.6. Mortality.** We identified only 4 fatalities associated with SC intake. MAM2201 (dose and route of administration unknown) was linked to the death of a 59-year-old Japanese man who was found dead at home (Saito et al., 2013) with MAM2201 detected in his femoral blood (1.24  $\mu\text{g/L}$ ), brain, body organs, and adipose tissues. Because there were no signs of physical injury and the deceased was assumed healthy, MAM2201 intoxication was considered cause of death.

In Sweden, a 17-year-old man was found alone outside (6–8 °C ambient temperature), dead from hypothermia and acute SC intoxication (Kronstrand et al., 2013). Prior to the man’s death, his friend reported smoking a foil of herb with the deceased. The friend took two whiffs, became light-headed and felt numbness in his hands. The friend went indoors afterwards, while the deceased continued smoking outside. JWH-210 was found in post-mortem femoral blood (12.3  $\mu\text{g/L}$ ).

A 26-year old man was found dead in his apartment with several bags containing methoxetamine (Wikström et al., 2013). Femoral blood concentration of 8.6  $\mu\text{g/kg}$  methoxetamine and three SC (AM694, AM2201, and JWH-018, all <1 pg/g) were confirmed. Although death was attributed to methoxetamine overdose, SC presence also may have contributed.

A 23-year-old man died from self-inflicted injuries sustained during a violent severe psychosis episode after smoking AM2201 (Patton et al., 2013). Prior to his death, a family member heard “stomping noises” for 30 min coming from his room. The man was eventually found dead on the floor with multiple injuries, including a fatal stab wound to his neck. A bag of “Mad Hatter” incense, smoke pipe, and a bag of white pills (labeled “ZAN-X”) were found in his room. AM2201 (12.0  $\mu\text{g/L}$ ) was identified in post-mortem heart blood. No other drugs were found. AM2201 also was detected in the “Mad Hatter” incense” and pipe residue. Traces of JWH-073 also were detected. “ZAN-X” did not contain any illicit or prescription drugs.

Reported SC blood concentrations in these cases were 1.2–12.3  $\mu\text{g/L}$  compared to impaired driver concentrations of 0.1–28  $\mu\text{g/L}$ . This overlap between lethal and DUID SC concentrations precludes ready identification of a fatal SC concentration. Other factors also could have contributed to death, such as undetected additional SC and/or other drugs of abuse, dose and route of administration, individual variation in SC metabolism, and lack of drug tolerance.

In summary, SC intake produced physiological and psychological effects in humans, which were qualitatively similar but of greater magnitude and duration than cannabis’ effects. Acute adverse effects generally subsided within 24–48 h, with patients treated with benzodiazepines and supportive care. Rarely, SC intake also caused acute kidney injury and death. In DUID suspects, SC blood concentrations were 0.7–28  $\mu\text{g/L}$ , with no clear correlation between impairment and blood concentrations. Withdrawal symptoms similar to those following chronic frequent cannabis intake were observed in chronic SC smokers after at least 1 week of abstinence.

**Table 3**  
Human synthetic cannabinoid (SC) administration studies.

| Country | N      | Sex    | Age      | SC brand                                      | Dose (route)                                | Symptoms                                                                                                                                                                                                                                                                | Time onset                  | Duration | Laboratory results (matrix)                                                                                                                                       | Reference                                |
|---------|--------|--------|----------|-----------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Germany | 2      | M      | –        | Spice                                         | 300 mg (sm)                                 | Reddened conjunctiva, dry mouth, elevated pulse, perception & mood change, mild impairment                                                                                                                                                                              | 10 min                      | 6–24 h   | (+)CP47,497 (blood)                                                                                                                                               | <a href="#">Auwärter et al. (2009)</a>   |
| Germany | 1      | M      | 47       | Smoke                                         | 150 mg (sm)                                 | Nausea, sedation, dry mouth, “hot flushes”, “burning eyes”, “thought disruption”, elevated pulse & blood pressure, slight pupil dilation, mild exhaustion                                                                                                               | “immediate”                 | 6–12 h   | (+)JWH-018 (serum) 8.1–10.2 µg/L after 5 min, <1 µg/L after 3 h                                                                                                   | <a href="#">Teske et al. (2010)</a>      |
| USA     | 1<br>6 | F<br>M | –        | K2 Standard (2), K2 Citron (2), K2 Summit (2) | 100 mg (sm)<br>300 mg (sm) one or two puffs | Reddened conjunctiva (6), burning eyes (1), xerostomia (4), tachycardia (6), anxiety (4), paranoia (2), sedation (4), mood & perception changes (5), thought disruption/loss of concentration (4), impaired sense of time (4), exhaustion (3), perceived impairment (5) | 2–3 min; peaked at 5–10 min | NS       | (+)4–5 µg/L JWH-018 & JWH-073 (blood) peak; <1 µg/L after 2 h                                                                                                     | <a href="#">Logan et al. (2011)</a>      |
| Russia  | 1      | M      | 47       | AM694                                         | 10 mg (po), 1 mg (sm)                       | None                                                                                                                                                                                                                                                                    | NS                          | NS       | (+)AM694 hydroxylated & pentanoic metabolites (urine)                                                                                                             | <a href="#">Grigoryev et al. (2012a)</a> |
| Russia  | 1      | M      | 47       | AB-001                                        | 13 mg (po)                                  | None                                                                                                                                                                                                                                                                    | NS                          | NS       | (+)AB-001 hydroxylated metabolites (urine) peaked after 5–7 h                                                                                                     | <a href="#">Grigoryev et al. (2012b)</a> |
| Germany | 1<br>1 | F<br>M | 43<br>42 | AM2201                                        | 26 mg (po)<br>5 mg (po)                     | None                                                                                                                                                                                                                                                                    | NS                          | NS       | (+)0.6 µg/L AM2201 at 1.5 h (serum); <0001 µg/L at 5d; AM2201 metabolites 0.05–0.2 µg/L & JWH-018 metabolites 0.7–0.3 µg/L (serum) at 1.5 h; metabolites in urine | <a href="#">Hutter et al. (2013)</a>     |

*Abbreviations:* AB-001 [(1-Pentyl-1H-indol-3-yl)adamantylmethanone], AM694 [1-(5-fluoropentyl)-3-(2-iodobenzoyl)indole], AM2201 [N-(5-Fluoropentyl)-3-(1-naphthoyl)indole], CP47,497 [2-[(1R,3S)-3-hydroxycyclohexyl]-5-(2-methyloctan-2-yl)phenol], JWH-018 [naphthalen-1-yl-(1-pentylindol-3-yl)methanone], JWH-073 [naphthalen-1-yl-(1-butyndol-3-yl)methanone], NS (not specified), po (oral administration), sm (smoked administration).

### 3.4. Receptor affinity

We identified 42 peer-reviewed journal articles on the interaction of SC with cannabinoid receptors; the majority were *in vitro* receptor binding studies and functional assays. SC psychoactive properties are attributed to CB<sub>1</sub> receptor activation (Pertwee, 2004). SC binding affinities were studied in a variety of *in vitro* models and further assessed with *in vivo* studies (*i.e.*, animal administration).

**3.4.1. Binding affinity at cannabinoid receptors.** Most cannabinoid receptor competitive binding affinity studies in rodent brain employed tritium [<sup>3</sup>H]-labeled CP55,940 or WIN55,212-2 (Pertwee, 2010) to evaluate receptor binding. Data were presented as the SC concentration displacing 50% of the radiolabeled compound from the receptor (IC<sub>50</sub>), which is utilized to calculate the binding affinity, K<sub>i</sub> (lower K<sub>i</sub> represents greater affinity). In general, SC binding affinity to CB<sub>1</sub> and/or CB<sub>2</sub> was compared with THC affinity. THC CB<sub>1</sub> and CB<sub>2</sub> binding affinities were 40.7 and 36.4 nM, respectively (Showalter et al., 1996).

The majority of SC detected in herbal products possessed higher affinity and lower K<sub>i</sub> values than THC at the CB<sub>1</sub> receptor (Table 4). JWH-018 metabolites retained CB<sub>1</sub> binding affinity

(*in vitro*) in mouse brain homogenates expressing CB<sub>1</sub> receptors (Brents et al., 2011). JWH-018, and its 4- and 5-hydroxyindole metabolites had K<sub>i</sub> values of 8.0, 2.6 ± 0.3 and 4.2 ± 1.2 nM, respectively, 5–6 times lower than THC's K<sub>i</sub> (15.3 ± 4.5 nM) in this preparation. JWH-018 6-hydroxyindole, 7-hydroxyindole and N-5-hydroxypentyl had equipotent (17.2 ± 3.6, 20.8 ± 4.9, and 25.2 ± 5.0 nM) binding affinities to THC. However, JWH-018 N-pentanoic acid CB<sub>1</sub> K<sub>i</sub> was >10,000 nM. JWH-018 4-hydroxyindole 10 mg/kg i.p. produced cannabinoid tetrad in mouse much greater than THC (Brents et al., 2011). Thus, some SC metabolites may be active and prolong the parent compound's psychoactive and physiological effects and contribute to intoxication severity.

SC receptor binding affinities and selectivity vary based upon functional group substitution (Aung et al., 2000; Wiley et al., 2013b). For example, JWH-018 (naphthoylindole with a N-pentyl side chain) and its N-butyl analog (JWH-073) CB<sub>1</sub> K<sub>i</sub> values are 9.0 ± 5.0 and 8.9 ± 1.8 nM, respectively (Aung et al., 2000). However, AM2201 (JWH-018 5-fluoropentyl analog) has a CB<sub>1</sub> K<sub>i</sub> of 1 nM (Makriyannis and Deng, 2000). This single halogenated analog is frequently encountered in newly emerging SC. Table 4 summarizes published SC binding affinities at CB<sub>1</sub> and CB<sub>2</sub> receptors.

**Table 4**  
Synthetic cannabinoid (SC) CB<sub>1</sub> and/or CB<sub>2</sub> receptor binding affinity.

| Compound     | Class                           | CB <sub>1</sub> K <sub>i</sub> (nM) | CB <sub>2</sub> K <sub>i</sub> (nM) | THC/SC CB <sub>1</sub> K <sub>i</sub> <sup>a</sup> | Reference                   |
|--------------|---------------------------------|-------------------------------------|-------------------------------------|----------------------------------------------------|-----------------------------|
| AB-FUBINACA  | Indazole carboxamide            | 0.9                                 | –                                   | 45.6                                               | Buchler et al. (2009)       |
| ADB-FUBINACA | Indazole carboxamide            | 0.4                                 | –                                   | 103                                                | Buchler et al. (2009)       |
| AM679        | Benzoylindoles                  | 13.5                                | 49.5                                | 3.0                                                | Makriyannis and Deng (2000) |
| AM694        | Benzoylindoles                  | 0.1                                 | 1.4                                 | 410                                                | Makriyannis and Deng (2000) |
| AM1220       | Naphthoylindoles                | 3.9                                 | 73.4                                | 10.5                                               | Makriyannis and Deng (2000) |
| AM1248       | Adamantylindoles                | 11.9                                | 4.8                                 | 3.4                                                | Makriyannis and Deng (2007) |
| AM2201       | Naphthoylindoles                | 1.0                                 | 2.6                                 | 40                                                 | Makriyannis and Deng (2000) |
| AM2232       | Naphthoylindoles                | 0.3                                 | 1.5                                 | 136                                                | Makriyannis and Deng (2000) |
| AM2233       | Naphthoylindoles                | 2.8                                 | –                                   | 14.6                                               | Deng et al. (2005)          |
| (±)CP47,497  | Cyclohexylphenols               | 2.2 ± 0.5                           | –                                   | 18.6                                               | Melvin et al. (1993)        |
| CP47,497(C8) | Cyclohexylphenols               | 0.8 ± 0.1                           | –                                   | 51.3                                               | Melvin et al. (1993)        |
| CP55,940     | Cyclohexylphenols               | 1.1 ± 0.04                          | –                                   | 37.3                                               | Melvin et al. (1993)        |
| HU-210       | Dibenzopyrans                   | 0.2                                 | 0.4 ± 0.1                           | 205                                                | Devane et al. (1992)        |
| JWH-015      | Naphthoylindoles                | 336 ± 36                            | 13.8 ± 4.6                          | 0.1                                                | Aung et al. (2000)          |
| JWH-018      | Naphthoylindoles                | 9.0 ± 5.0                           | 2.9 ± 2.7                           | 4.6                                                | Aung et al. (2000)          |
| JWH-019      | Naphthoylindoles                | 9.8 ± 2.0                           | 5.6 ± 2.0                           | 4.2                                                | Aung et al. (2000)          |
| JWH-030      | Naphthoylindoles                | 87.0 ± 3.0                          | –                                   | 0.5                                                | Tarzia et al. (2003)        |
| JWH-073      | Naphthoylindoles                | 8.9 ± 1.8                           | 38.0 ± 24.0                         | 4.6                                                | Aung et al. (2000)          |
| JWH-081      | Naphthoylindoles                | 1.2 ± 0.03                          | 12.4 ± 2.2                          | 34.2                                               | Aung et al. (2000)          |
| JWH-122      | Naphthoylindoles                | 0.7 ± 0.5                           | 1.2 ± 1.2                           | 58.6                                               | Huffman et al. (2003)       |
| JWH-200      | Naphthoylindoles                | 42.0 ± 5.0                          | –                                   | 1.0                                                | Huffman et al. (2003)       |
| JWH-203      | Naphthoylindoles                | 8.0 ± 0.9                           | 7.0 ± 1.3                           | 5.1                                                | Huffman et al. (2005a)      |
| JWH-210      | Naphthoylindoles                | 0.5 ± 0.03                          | 0.7 ± 0.01                          | 82                                                 | Huffman et al. (2005b)      |
| JWH-250      | Phenylacetylindoles             | 11.0 ± 2.0                          | 33.0 ± 2.0                          | 3.7                                                | Huffman et al. (2005a)      |
| JWH-251      | Phenylacetylindoles             | 29.0 ± 3.0                          | 146.0 ± 36                          | 1.4                                                | Huffman et al. (2005a)      |
| JWH-307      | Naphthoylpyrroles               | 7.7                                 | –                                   | 5.3                                                | Huffman et al. (2005a)      |
| THC          | Dibenzopyran                    | 41 ± 2                              | 36 ± 10                             | 1.0                                                | Showalter et al. (1996)     |
| UR-144       | Tetramethylcyclo-propyl indoles | 29.0 ± 0.9                          | 4.5 ± 1.7                           | 1.4                                                | Wiley et al. (2013a)        |
| WIN55,212-2  | Aminoalkylindoles               | 62.3                                | 3.3                                 | 0.7                                                | Felder et al. (1995)        |
| XLR11        | Tetramethylcyclo-propyl indoles | 24.0 ± 4.6                          | 2.1 ± 0.6                           | 1.7                                                | Wiley et al. (2013a)        |

**Abbreviations:** AB-FUBINACA [N-[(2S)-1-Amino-3-methyl-1-oxo-2-butanyl]-1-(4-fluorobenzyl)-1H-indazole-3-carboxamide], ADB-FUBINACA [N-(1-Amino-3,3-dimethyl-1-oxobutan-2-yl)-1-(4-fluorobenzyl)-1H-indazole-3-carboxamide], AM679 [1-Pentyl-3-(2-iodo-benzoyl)indole], AM694 [1-(5-Fluoropentyl)-3-(2-iodobenzoyl)indole], AM1220 [1-[(N-Methylpiperidin-2-yl)methyl]-3-(1-naphthoyl)indole], AM1248 [1-[(1-Methyl-2-piperidin-yl)methyl]-1H-indol-3-yl]-adamantylmethanone, AM2201 [N-(5-fluoropentyl)-3-(1-naphthoyl)indole], AM2232 [5-(3-(1-Naphthoyl)-1H-indol-1-yl)pentannitrile], AM2233 [(2-Iodophenyl)-[1-[(1-methyl-2-piperidinyl)methyl]-1H-indol-3-yl]methanone], CP47,497 [2-[(1S,3S)-3-Hydroxycyclohexyl]-5-(2-methyloctan-2-yl)phenol], CP47,497(C8) [trans-3-(4-(1,1-Dimethyloctyl)-2-hydroxyphenyl)cyclohexanol], HU-210 [(6aR)-trans-3-(1,1-Dimethylheptyl)-6a,7,10,10a-tetrahydro-1-hydroxy-6,6-dimethyl-6Hdibenz[b,d]pyran-9-methanol], JWH-015 [2-Methyl-1-propyl-3-(1-naphthoyl)indole], JWH-018 [Naphthalen-1-yl(1-pentyl-indol-3-yl)methanone], JWH-019 [N-Hexyl-3-(1-naphthoyl)indole], JWH-030 [3-(1-Naphthoyl)-1-pentylpyrrole], JWH-073 [N-Butyl-3-(1-naphthoyl)indole], JWH-081 [N-Pentyl-3-[1-(4-methoxy)-naphthoyl]indole], JWH-122 [1-Pentyl-3-(1-(4-methylnaphthoyl)indole], JWH-200 [1-(2-(Morpholin-4-yl)ethyl)-3-(1-naphthoyl)indole], JWH-203 [1-Pentyl-3-(2-chlorophenylacetyl)indole], JWH-210 [1-Pentyl-3-(1-(4-ethylnaphthoyl)indole], JWH-250 [1-Pentyl-3-(2-methoxy-phenylacetyl)indole], JWH-251 [2-(2-Methylphenyl)-1-(1-pentyl-1H-indol-3-yl)-ethanone], JWH-307 [5-[(2-Fluorophenyl)-1-pentylpyrrol-3-yl]-1 (naphthalenyl)methanone]; THC (delta-9-tetrahydrocannabinol), UR-144 [1-Pentylindol-3-yl)-(2,2,3,3-tetramethylcyclopropyl)methanone], XLR11[1-(5-Fluoropentyl)-1H-indol-3-yl](2,2,3,3-tetramethylcyclopropyl)methanone].

<sup>a</sup> Ratio calculated between THC CB<sub>1</sub> K<sub>i</sub> = 41 nM and mean CB<sub>1</sub> K<sub>i</sub> for each synthetic cannabinoid. Ratio >1 represents greater CB<sub>1</sub> binding affinity than for THC.



**3.4.2. Intrinsic activity at cannabinoid receptors.**  $K_i$  presents only ligand-binding interaction. *In vitro* or *in vivo* functional assays, which evaluate the drug's efficacy (e.g.,  $EC_{50}$ ) and potency, are needed to determine a drug's intrinsic activity as a receptor ligand, i.e., full, partial, or inverse agonist, or neutral antagonist. However, we identified few published data on SC efficacy or intrinsic activity. SC intrinsic activities were studied in a variety of *in vitro* models, making direct comparisons challenging.  $CB_1$  agonist activation down regulates adenylyl cyclase and decreases cellular cAMP levels, which triggers a cascade of reactions that affects cellular signaling and neurotransmitter inhibition, including acetylcholine, dopamine, noradrenaline, and glutamine, and  $\gamma$ -aminobutyric acid (GABA) (Mechoulam and Parker, 2013; Pertwee, 2010).

Reduced cellular cAMP levels also were observed with  $CB_2$  activation (Slipetz et al., 1995). Moreover,  $CB_1$  or  $CB_2$  receptor activation was measured with [ $^{35}$ S]GTP $\gamma$ S functional assay, which involved G-protein conformational change (Breivogel et al., 1997).

CP55,940 inhibited adenylyl cyclase production more effectively at 25 nM compared to THC (430 nM) (Howlett et al., 1988). CP55,940, HU-210, and WIN55,212-2 concentration-dependent inhibited forskolin-induced cAMP production by activating  $CB_2$  receptors expressed in Chinese hamster ovarian cells, which were blocked by pertussis toxin at 10 ng/mL maximal inhibition (Slipetz et al., 1995).

HU-210, CP55,940, and WIN55,212-2 activated  $CB_1$  receptors in a [ $^{35}$ S]GTP $\gamma$ S binding assay with a potency order of: HU-210 > CP55,940 > THC > WIN55,212-2 (Breivogel et al., 2001). JWH-018 and its mono-hydroxylated metabolites produced partial G-protein activation in a mouse  $CB_1$  receptor preparation with greater potency than THC (Brents et al., 2011). Naphthoylindoles identified in seized herbal products activated  $CB_1$  receptors (Nakajima et al., 2011). Tetramethylpropylindoles, XLR-11 and UR-144 also were reported to have  $CB_1$  and  $CB_2$  agonist activities (Wiley et al., 2013a).

$CB_1$  receptors activated by JWH-018, JWH-073 and CP 47,497 inhibited release of neurotransmitters in autaptic neurons (isolated cultured neurons that form synapses onto themselves) and inhibition was reversed by rimonabant treatment (Atwood et al., 2011). Table 5 summarizes SC  $CB_1$  and  $CB_2$  intrinsic activity.

SC intrinsic activities appear to have full or partial agonist properties similar to that of THC; though SC produce *in vivo* pharmacodynamic effects at lower median effective dose ( $ED_{50}$ ), which is the drug dose required to produce measurable response in 50% of the test population.

In summary, SC binding affinities to  $CB_1$  and  $CB_2$  receptors had varying intrinsic values when measured by *in vitro* functional assays, but  $K_i$  values were not necessarily equivalent to  $ED_{50}$  in animals. The majority of SC found in seized materials have unknown

**Table 5**  
Agonist-stimulated activation of  $CB_1$  and  $CB_2$  receptors by synthetic cannabinoids represented as median maximum effective dose ( $EC_{50}$ ) or inhibitor binding affinity ( $K_i$ ) values.

| Drug                    | Model                       | $EC_{50}$ or $K_i$ (nM)      |                | Intrinsic activity at $CB_1$ receptor | Reference                                        |
|-------------------------|-----------------------------|------------------------------|----------------|---------------------------------------|--------------------------------------------------|
|                         |                             | $CB_1$                       | $CB_2$         |                                       |                                                  |
| AB-FUBINACA             | GTP $\gamma$ S binding      | 23.2                         | –              | Full agonist                          | Buchler et al. (2009)                            |
| ADB-FUBINACA            | GTP $\gamma$ S binding      | 1.0                          | –              | Full agonist                          | Buchler et al. (2009)                            |
| AM2201                  | GTP $\gamma$ S binding      | 24.4                         | –              | Full agonist                          | Nakajima et al. (2011)                           |
| AM694                   | GTP $\gamma$ S binding      | 52.8                         | –              | Full agonist                          | Nakajima et al. (2011)                           |
| CP47,497 C8             | Internalization assay       | 4.4                          | –              | Full agonist                          | Atwood et al. (2011)                             |
| (–)CP55,940             | adenylyl cyclase inhibition | 25                           | 0.7 $\pm$ 0.2  | Full agonist                          | Howlett et al. (1988) and Slipetz et al. (1995)  |
| CP55,940                | GTP $\gamma$ S binding      | 3.4 $\pm$ 2.4, 25 $\pm$ 14.4 | 23 $\pm$ 6.8   | Full agonist                          | Brents et al. (2011) and Wiley et al. (2013a)    |
| HU-210                  | adenylyl cyclase inhibition | –                            | 0.4 $\pm$ .07  | –                                     | Slipetz et al. (1995)                            |
| HU-210                  | GTP $\gamma$ S binding      | 2.9                          | –              | Full agonist                          | Breivogel et al. (2001)                          |
| JWH-018                 | Internalization assay       | 10.1                         | –              | Full agonist                          | Atwood et al. (2011)                             |
| JWH-018                 | GTP $\gamma$ S binding      | 36, 6.8 $\pm$ 2.5            | –              | Partial to full agonist               | Brents et al. (2011) and Nakajima et al. (2011)  |
| JWH-018 4-hydroxyindole | GTP $\gamma$ S binding      | 17.0 $\pm$ 9.6               | –              | Partial to full agonist               | Brents et al. (2011)                             |
| JWH-019                 | GTP $\gamma$ S binding      | 98.9                         | –              | Full agonist                          | Nakajima et al. (2011)                           |
| JWH-073                 | Internalization assay       | 45.6                         | –              | Full agonist                          | Atwood et al. (2011)                             |
| JWH-210                 | GTP $\gamma$ S binding      | 20.4                         | –              | Full agonist                          | Nakajima et al. (2011)                           |
| JWH-251                 | GTP $\gamma$ S binding      | 29.0 $\pm$ 5.5               | 8.3 $\pm$ 0.8  | Full agonist                          | Huffman et al. (2005a)                           |
| JWH-302                 | GTP $\gamma$ S binding      | 29.3 $\pm$ 0.8               | 24.4 $\pm$ 6.9 | Full agonist                          | Huffman et al. (2005a)                           |
| RCS-4                   | GTP $\gamma$ S binding      | 199                          | –              | Partial agonist                       | Nakajima et al. (2011)                           |
| THC                     | adenylyl cyclase inhibition | 430                          | –              | Partial agonist                       | Dill and Howlett (1988)                          |
| THC                     | GTP $\gamma$ S binding      | 81 $\pm$ 34, 167 $\pm$ 4.7   | –              | Partial agonist to full agonist       | Breivogel et al. (2001) and Brents et al. (2011) |
| UR-144                  | GTP $\gamma$ S binding      | 95 $\pm$ 20                  | 334 $\pm$ 170  | Full agonist                          | Wiley et al. (2013a)                             |
| XLR-11                  | GTP $\gamma$ S binding      | 159 $\pm$ 38                 | 145 $\pm$ 73.8 | Full agonist                          | Wiley et al. (2013a)                             |
| WIN55,212-2             | adenylyl cyclase inhibition | 24 $\pm$ 3.7                 | 0.7 $\pm$ 0.2  | Full agonist                          | Felder et al. (1992) and Slipetz et al. (1995)   |
| WIN55,212-2             | GTP $\gamma$ S binding      | 170 $\pm$ 80                 | –              | Full agonist                          | Breivogel et al. (2001)                          |

**Abbreviations:** AB-FUBINACA [N-(2S)-1-Amino-3-methyl-1-oxo-2-butanyl]-1-(4-fluorobenzyl)-1H-indazole-3-carboxamide], ADB-FUBINACA [N-(1-Amino-3,3-dimethyl-1-oxobutan-2-yl)-1-4-fluorobenzyl-1H-indazole-3-carboxamide], AM694 [1-(5-Fluoropentyl)-3-(2-iodobenzoyl)indole], AM2201 [N-(5-fluoropentyl)-3-1-(naphthylindole)], CP47,497(C8) [trans-3-(4-(1,1-Dimethyloctyl)-2-hydroxyphenyl)cyclohexanol], HU-210 [(6aR)-trans-3-(1,1-Dimethylheptyl)-6a,7,10,10a-tetrahydro-1-hydroxy-6,6-dimethyl-6Hdibenzo[b,d]pyran-9-methanol], JWH-015 [2-Methyl-1-propyl-3-(1-naphthoyl)indole], JWH-018 [Naphthalen-1-yl(1-pentyl-indol-3-yl)methanone], JWH-019 [N-Hexyl-3-(1-naphthoyl)indole], JWH-030 [3-(1-Naphthoyl)-1-pentylpyrrole], JWH-073 [N-Butyl-3-(1-naphthoyl)indole], JWH-081 [N-Pentyl-3-(1-(4-methoxy)-naphthoyl)indole], JWH-122 [1-Pentyl-3-(1-(4-methylnaphthoyl)indole], JWH-200 [1-(2-(Morpholin-4-yl)ethyl)-3-(1-naphthoyl)indole], JWH-203 [1-Pentyl-3-(2-chlorophenylacetyl)indole], JWH-210 [1-Pentyl-3-(1-(4-ethylnaphthoyl)indole], JWH-250 [1-Pentyl-3-(2-methoxy-phenylacetyl)indole], JWH-251 [2-(2-Methylphenyl)-1-(1-pentyl-1H-indol-3-yl)-ethanone], JWH-307 [5-[(2-Fluorophenyl)-1-pentylpyrrol-3-yl]-1-(naphthalenyl)methanone], UR-144 [(1-Pentylindol-3-yl)-(2,2,3,3-tetramethylcyclopropyl)methanone], XLR11 [(1-(5-Fluoropentyl)-1H-indol-3-yl)(2,2,3,3-tetramethyl-cyclopropyl)-methanone], WIN55,212-2 [(R)-(+)-[2,3-Dihydro-5-methyl-3-(4-morpholinylmethyl)pyrrolo[1,2,3-de]-1,4-benzoxazin-6-yl]-1-naphthalen-ylmethanone].



intrinsic values. *In vivo* evaluation of intrinsic activities could assist clinical treatment development for SC-related symptoms.

## 4. Conclusion

### 4.1. Summary

SC consumption has become widespread, despite law enforcement and regulatory control measures. Epidemiological data suggest that the majority of SC users are young adults who perceive SC as safer than non-cannabinoid illicit drugs and a favorable cannabis alternative eliciting cannabis-like “high” while avoiding detection by standard drug screens. However, data suggest that many SC users prefer cannabis over SC due to the drugs’ negative effects. SC are readily accessible, sold under several names and packaging with smoking as the most common route of administration. Most SC smokers are men from 13 to 59 years old, many with a history of polydrug use such as cannabis, alcohol, and nicotine.

SC were investigated in animals to characterize THC effects and evaluate their therapeutic benefits. Acute SC administration in rodents produced the cannabinoid tetrad of effects with dose-dependent anxiolytic and anxiogenic properties. Chronic SC administration was anxiogenic and produced more pronounced behavioral deficits in rodents exposed during adolescence, but no significant physiological effects in animals exposed *in utero*. SC substituted for THC in animal (rats and monkeys) discrimination studies, and also attenuated antagonist-elicited withdrawal in monkeys previously exposed to THC.

Recreational SC intake arose in the 2000s and many adverse effects were reported. Acute SC intoxication can lead to ED presentation and hospitalization, requiring supportive care, benzodiazepines, and fluids. While most such patients were released within 24 h of admission, severe adverse effects such as cardiotoxicity, AKI, and psychosis resulted in hospitalization for as long as 2 weeks. Deaths directly linked to SC use were quite rare. Some chronic SC users experienced withdrawal symptoms when they stopped drug intake.

Most SC have greater binding affinity to CB<sub>1</sub> receptors than does THC, suggesting a possible mechanism for the severity of acute clinical reactions that result in ED presentation. However, SC intrinsic activity data are limited, with few direct comparisons to THC, making it premature to draw any conclusions about mechanisms.

### 4.2. Knowledge gaps and limitations

SC epidemiology data are limited and derived mostly from cross-sectional surveys of small, self-selected convenience samples. Community-based epidemiological surveys with large, nationally representative samples are needed to better understand SC epidemiology.

Although hundreds of animal studies were conducted with SC, the majority focused on CP55,940, HU-210, and WIN55-212-2. Newer SC identified in herbal products and human clinical samples need further investigation.

Human controlled SC administration studies and systematic *in vitro* and *in vivo* pharmacokinetic studies are needed to fill in important gaps in our knowledge of SC pharmacokinetics, tissue/organ distribution, elimination, metabolite biological activity, and drug–drug interactions. Human studies will be difficult to conduct in the US, as they must go through a rigorous multi-agency approval process (*i.e.*, Drug Enforcement Administration, Food and Drug Administration, and Institutional Review Board) before they can be conducted; the preclinical safety and pharmacokinetic data to support such applications are lacking.

Analytical laboratories are challenged with SC identification in biological matrices due to structural diversity and similarity. Unidentified SC in a patient’s sample, makes it difficult to definitively evaluate SC clinical effects or develop specific treatments. Medical treatment must often remain symptomatic. To address the increasing safety and public health issues associated with SC intake, future studies are needed to evaluate SC abuse liability, *in utero* and long-term effects, which are part of developing withdrawal and addiction treatment similar to the effects elicited by cannabis.

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### Contributors

Marisol Castaneto, David Gorelick, Nathalie Desrosiers, and Rebecca Hartman reviewed and selected articles as part of the comprehensive literature review. Marisol Castaneto organized the data included in this review and wrote the first draft of the manuscript with the assistance of David Gorelick, Sandrine Pirard, and Marilyn Huestis. All authors contributed and approved the final manuscript.

### Conflict of interest

No conflict declared.

### Disclaimer

The opinions or assertions herein are those of the authors and do not necessarily reflect the views of the NIH or the Departments of the Army, Navy, or the Department of Defense.

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### Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.drugalcdep.2014.08.005>.

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